Introduction to Quantum Mechanics

Joseph D. MacMillan



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Preface

To come.

Why Open Education?

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Why did I choose this? Because I've taught university courses and have seen the issues with expensive textbooks: students not being able to afford them, students finding illegal PDF copies online and sharing them, the publishing companies fighting back with various schemes like renting digital copes, and so on. Especially in advanced physics, little textbooks can be so expensive, and yet I think textbooks are an important and necessary resource, and I want my students to all have a copy.

Thanks

To come.

Find a typo, mistake, or horrible misconception in this book? Let me know by creating a new issue at the GitHub page:

github.com/josephmacmillan/IntroductionToQuantumMechanics/issues.

Part I

Quantum Mechanics in Zero Dimensions

Chapter 1

The Stern-Gerlach Experiment

1.1 Spin Angular Momentum

Let's start with something that is pretty much as far away from the quantum world as possible – the Earth and Sun. With its total motion, the Earth has two kinds of angular momentum: *orbital* angular momentum $\mathbf{L}$ from its orbit around the sun, and *spin* angular momentum $\mathbf{S}$, resulting from its 24 hour rotation. Of course, the spin angular momentum in this case is due to adding up the orbital angular momenta $\mathbf{L}_i = \mathbf{r}_i \times \mathbf{p}_i$ of each of the tiny pieces of the Earth, but we'll keep the distinction – it'll turn out that this is *not* the case in quantum systems.[1]

The quantum version of this kind of system could be an electron in orbit around a proton – a hydrogen atom. The electron has orbital angular momentum just like the Earth (depending on what state it's in), and it also has spin angular momentum. Careful, though, as the electron doesn't rotate like the Earth – how can it when it has essentially no size or diameter to spin? Despite this, it has measurable intrinsic angular momentum, which we'll call *spin* $\mathbf{S}$. Since spin is a vector, it has components (S_x, S_y, S_z), and thus to specify the spin of the electron we need to use three different numbers; keep this in mind for later.

I'd like to explore this idea of spin without the complication of orbital angular momentum, so let's put a stationary electron all by itself in a magnetic field $\mathbf{B}$. Since the electron isn't moving, the Lorentz force

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$$

[1] Note that the two kinds of angular momentum for the Earth point in different directions since the rotation axis of the Earth is tilted 23° from the orbital plane.

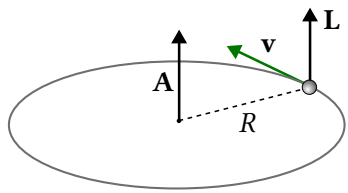


Figure 1.1: A charged particle in a circular orbit has a magnetic dipole moment.

is zero. But the electron's spin angular momentum gives it a magnetic dipole moment $\boldsymbol{\mu}$, and it's then possible for an *inhomogeneous* magnetic field to exert a force given by[2]

$$\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}). \tag{1.1}$$

Now, we can calculate the magnetic dipole moment due to orbital angular momentum easily enough (we'll handle spin in a moment); Figure 1.1 shows a particle in a circular orbit of radius R and circular velocity $\mathbf{v}$. If we think of this as a current loop, the dipole moment is just the current I times the area $\mathbf{A}$,

$$\boldsymbol{\mu} = I\mathbf{A}.$$

The magnitude of the area vector is just $A = \pi R^2$ and the direction is perpendicular to the orbit, while the current will be the charge divided by the time it takes for the charge to make the orbit:

$$I = \frac{q}{T} = \frac{q}{2\pi R/v}.$$

The dipole moment is then

$$\boldsymbol{\mu} = \left(\frac{1}{2}qvR, \text{ perpendicular to the orbit}\right).$$

It's more useful, though, to put this in terms of the angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, which for a circular orbit is

$$\mathbf{L} = (rmv, \text{ perpendicular to the orbit}).$$

Finally, then, the magnetic dipole moment arising from orbital angular momentum is

$$\boldsymbol{\mu} = \frac{q}{2m}\mathbf{L}. \tag{1.2}$$

[2]See Griffiths *Introduction to Electrodynamics*, fifth edition, section 6.1.2.

What changes for spin angular momentum? It turns out to be only slightly different, and we can write

$$\boldsymbol{\mu} = \frac{gq}{2m}\mathbf{S}, \tag{1.3}$$

where g is called the gyroscopic ratio or often just "g-factor" and, for an electron, is *almost* $g = 2$.[3]

Let's go back to our electron in an inhomogeneous field. Take the field to be constant in the x and y directions but *not* in the z direction. Then

$$\boldsymbol{\mu} \cdot \mathbf{B} = \mu_x B_x + \mu_y B_y + \mu_z B_z,$$

and since the first two terms are constant, equation (1.1) gives

$$\mathbf{F} = \nabla(\mu_z B_z) = \mu_z \frac{\partial B_z}{\partial z}\,\hat{z} = \frac{gq}{2m}\frac{\partial B_z}{\partial z} S_z\,\hat{z}. \tag{1.4}$$

The force on the electron in this case is directed along the z direction and proportional to the component of spin angular momentum S_z along that direction – the more the spin aligns with the z axis, the greater the force.

1.2 The Stern-Gerlach Experiment

In 1922 Otto Stern and Walther Gerlach used these ideas to measure the spin angular momentum of electrons. Their experiment is shown in Figure 1.2, and they actually used silver atoms rather than electrons. Why silver? The atoms are electrically neutral, so there won't be any direct magnetic force on the atoms from the magnetic field, and the electronic configuration of silver is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^{10} 5s^1$. Except for the outermost electron, all others are in closed shells, meaning we can ignore their contribution to the angular momentum of the silver atom. The outermost electron is in an s-orbit, which has no orbital angular momentum – so the only angular momentum of the entire silver atom is due to the spin of that one outer electron.

The Stern-Gerlach experiment proceeds as follows. First, the oven boils silver atoms off of a solid piece of silver; the atoms are "thermalized," meaning they come out in all directions and with random spin orientations. A collimating slit produces a narrow beam of these atoms which pass between the magnets. The magnets are constructed to create an inhomogeneous magnetic field, with the inhomogeneity in the z direction;

[3] We don't need to worry about this here, but each charged particle with spin has its own g-factor – the proton has $g = 5.58$, for example. These can be measured and in some cases calculated theoretically, but it requires quantum field theory to do it properly.

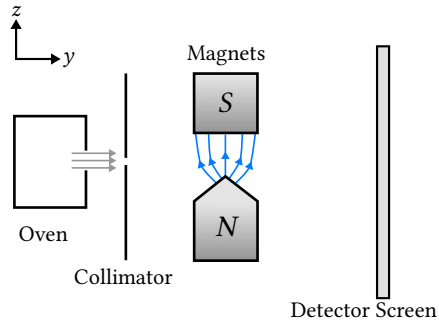


Figure 1.2: The Stern-Gerlach experiments, in which silver atoms are deflected by an inhomogeneous magnetic field.

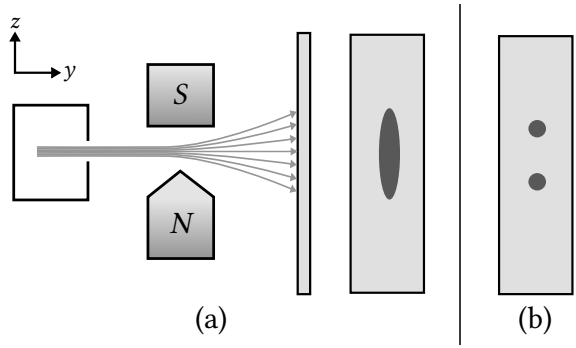


Figure 1.3: (a) Classically expected result. (b) Actual result showing quantization of spin angular momentum.

note that as shown the magnetic field gets stronger in the negative z direction, so $\partial B_z/\partial z$ is negative in equation (1.4). Because the charge on the electron is negative as well, the force on a silver atom passing through the field will be up along z if the spin has a positive z component, and down along $-z$ if the spin has a negative z component. Finally, the deflection of each atom is measured when they hit the detecting screen.

What do we expect? Well, as mentioned above, the oven thermalizes the atoms so the spin of the outer electron is in random directions. That means the force on each atom will be random as well – some atoms will get pushed up, some will get pushed down, and each with varying strength (up to some maximum for those spins that are entirely along z or $-z$). This expected result is shown in Figure 1.3 (a).

But that's not at all what Stern and Gerlach saw; instead, as Figure

1.3 (b) shows, they detected only two locations where silver atoms hit the screen, corresponding to spin angular momentum values of either

$$S_z = +\frac{\hbar}{2} \quad \text{or} \quad S_z = -\frac{\hbar}{2}.$$

This is clear evidence of the *quantization* of angular momentum, something Niels Bohr suggested in 1913 when developing his model of the atom. Evidently, every silver atom boiled off with the outermost electron having only one of two possible values for S_z. Of course, this says nothing about the values of S_x or S_y, or even the magnitude of the spin S. And, actually, it turns out we can't even say they had these values of S_z when they were created – we can only say we measured S_z to be either one or the other; more on this later.

1.3 Extending the Experiments

Before we discuss this further, though, I want to introduce some notation:

$$\boxed{|\psi\rangle \rightarrow \text{ Represents the state of a quantum mechanical system.}} \quad (1.5)$$

This notation is often called a "ket," for reasons we'll see later, and it includes all possible information you could know about a particular quantum system. Consider, for example, a silver atom that has been measured to have $S_z = +\hbar/2$. We could represent the state of the atom by

$$|+\rangle \rightarrow \text{ State for which } S_z = +\hbar/2 \text{ has been measured,}$$

although it's also common to use $|\uparrow\rangle$ or $|+z\rangle$. This state is usually called "spin up." Similarly, the "spin down" state is represented by

$$|-\rangle \rightarrow \text{ State for which } S_z = -\hbar/2 \text{ has been measured,}$$

or $|\downarrow\rangle$ or $|-z\rangle$. Any particle that, when measuring a component of its spin angular momentum, has only two possibilities for the results is called a *spin-1/2* particle; electrons, protons, and neutrons are all spin-1/2 particles.

Let's run through a few extensions to the original Stern-Gerlach experiment. I've simplified the diagram of the original experiment in Figure 1.4 to highlight the important steps: a beam of spin-1/2 particles (whether silver atoms or something else) is created, all in state $|\psi\rangle$, and sent to a Stern-Gerlach device (the magnets) to have the spin of each particle measured. In the case of the original Stern-Gerlach experiment, if a beam of

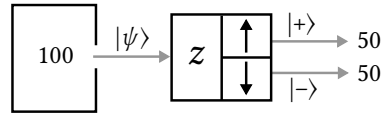


Figure 1.4: A simplified diagram of the original Stern-Gerlach experiment.

100 atoms were created, half of the atoms would be measured as spin up and half would be spin down.

Actually, we should be more careful – we can't predict which atoms will be spin up and which will be spin down, and the process is inherently *random*. That means that we might get 49 spin up and 51 spin down, or 43 and 57, but if we increase the number of particles in the beam, we'll get closer and closer to exactly 50% for each.

In these experiments, we can think of the Stern-Gerlach device as an *analyzer* – it measures something. But we can also think of it as a device for preparing a beam of particles all in the same known state; for example, each particle coming out of the top of the Stern-Gerlach device is in state $|+\rangle$, and we can use this knowledge to do further measurements.

1.3.1 Experiment 1

It may be a basic question, but it's worth asking: once the particles are in state $|+\rangle$, for example, do they *stay* in that state? In Experiment 1 we'll test this by connecting another Stern-Gerlach device as shown in Figure 1.5, and sure enough all particles that enter the second analyzer come out with the same $|+\rangle$ state, and none become spin down.

That's a good thing: it means we can do repeated measurements and expect the same results. Science would be tricky to do if things were otherwise. But we should be a little careful, as quantum systems do evolve in time – we'll learn about this later when we cover the Schrödinger equation – so the repeated measurement should be made right after the first.

What if we moved the second device to accept the spin down particles from the first? As you probably would guess, those particles would all be measured as being spin down, with no spin up particles at all.

1.3.2 Experiment 2

Now, we've been aligning the magnets in the Stern-Gerlach device so that the inhomogeneity in the magnetic field is along the z axis, but we could have just as easily rotated it (or changed our axes) so that it is in fact along

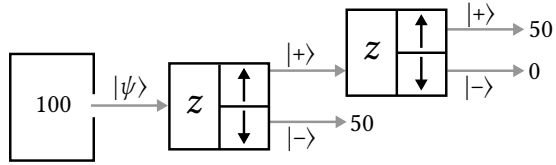


Figure 1.5: Experiment 1: the spin up particles are sent to a second analyzer, where they're found to still be spin up.

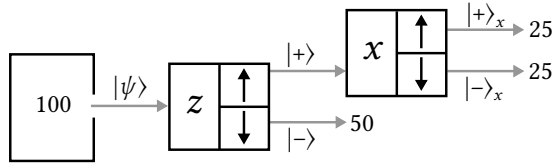


Figure 1.6: Experiment 2: the spin up particles are sent to a second analyzer that has been rotated so that it measures S_x.

the x direction. Does anything change? Well, no – the coordinate system is just our choice and can't affect the actual physics that's happening, so we could just replace every z in the discussion above with an x with no other change. But let's try something a little more interesting and combine a Stern-Gerlach device oriented along z with one oriented along x; this is shown in Figure 1.6.

First, note the new notation; a particle in the state $|+\rangle_x$ has had its component of spin angular momentum along the x direction measured to be $S_x = +\hbar/2$. This is a different state that our previous spin up state (you might call it "spin up along x" or something similar) and requires new notation. Note that I didn't bother with the z subscript for our S_z states; by convention, a ket without a subscript indicating direction will be along z.

The results are different this time, with half of the spin up particles sent into the second Stern-Gerlach device being spin up along x (so in state $|+\rangle_x$) and half being spin down along x (in state $|-\rangle_x$).

Remember what we're measuring here: the oven creates particles with a random spin orientation, and we measure half of those particles to have $S_z = +\hbar/2$, and then half of *those* particles to have $S_x = +\hbar/2$. We're missing the component of spin along the y direction, so you might think we just add a third device to measure S_y – and then we'd know all

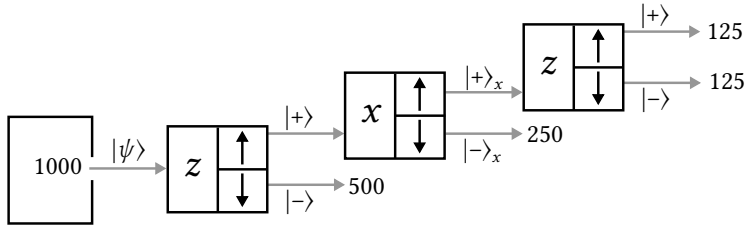


Figure 1.7: Experiment 3: the spin up particles are sent to a second analyzer that has been rotated so that it measures S_x, and then the particles in state $|+\rangle_x$ are sent to a third analyzer to have their spin component S_z measured again.

three components of spin and therefore the vector **S**. But there's an interesting twist to this simple story that the next experiment will highlight.

1.3.3 Experiment 3

So, in Experiment 2, spin up (along z) particles have their spin along x measured, and half are found to be spin up along x. In Experiment 3, we take those particles that are spin up along both directions and send them into a third analyzer, this one oriented along z as shown in Figure 1.7. Just so that we're clear: none of the particles sent into this last analyzer were measured to have $S_z = -\hbar/2$. Despite this, half the particles analyzed by the last Stern-Gerlach device are found to be in state $|-\rangle$ along z.

What do we to make of these experimental results? Evidently measuring S_x has *changed* the state of the particles; although they *were* in state $|+\rangle$, after the measurement of S_x they aren't any more.

Some more terminology: we call quantities that we can measure *observables*, and we say that S_z and S_x are *incompatible observables* – measuring one affects our knowledge of the other. Two other more famous incompatible observables are position and momentum from Heisenberg's uncertainty principle, although not all observables are incompatible – you could know a particle's position and spin S_z at the same time, for example.

Obviously we could have used S_y instead with no change to the results, so we run into a big problem measuring angular momentum: we need three components to specify it, but can never know more than one at a time! Actually, it turns out we can also know the *magnitude* S (we'll see why later), so at most we can specify the length and one component of the angular momentum vector.

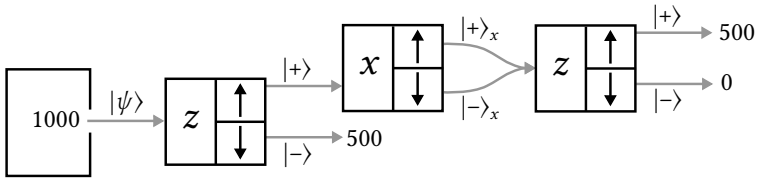


Figure 1.8: Experiment 4: We'll recombine the outputs from the S_x device without measuring anything and send that combined beam into an S_z device.

1.3.4 Experiment 4

For this last experiment, we need to make some changes to the Stern-Gerlach device measuring S_x. The details aren't too important as to how, but rather than measuring the spin S_x observable, we'll recombine the two beams – one pushed up by the magnetic field and one pushed down – and send that combined beam into the last Stern-Gerlach device aligned along the z direction. But now the results are different, and we find that every particle is spin up along z. It seems as though the middle device no longer has an effect on the initial $|+\rangle$ state of the particle; apparently it's the act of *measurement itself* that alters the state, not the push by the magnetic field. This is similar to the famous results from the double slit experiment – measuring which slit the particle goes through destroys the interference pattern.

Problems

Problem 1.1 – Electron spin. This whole chapter is about electron spin, but of course the electron doesn't really spin around like a top. But let's pretend for a moment that an electron really is a tiny sphere, of radius

$$r_e = \frac{e^2}{4\pi\epsilon_0 mc^2}$$

(the so-called classical electron radius), and that it spins around its axis with angular momentum $S_z = \hbar/2$. How fast would a point on the "equator" be moving? Does this model make sense?

Chapter 2

The Quantum State Vector

2.1 Basis States

In the previous chapter I introduced the ket $|\psi\rangle$, and said that it represents a *state* of a quantum mechanical system. But what exactly is it? It's hard to define, since by its nature it's abstract – in some ways, it's meant to represent everything we could possibly know about a particular quantum system. If we wanted to be technical I could tell you that it's a vector in complex Hilbert space, but I'm not sure that will help. Instead, I think it's best to draw an analogy with something you know well: vectors in three dimensional position space. Before we do that, though, I want to also introduce a *different* way of representing a quantum space, called (for reasons we'll see shortly) a "bra":

$$\boxed{\langle\psi| \rightarrow \text{ Represents the state of a quantum mechanical system.}} \quad (2.1)$$

Our job in this chapter is to get a good working idea of what a ket and bra actually are and how we use them in quantum mechanics.

I'm sure you recall some of the basics of how we write vectors – that is, normal old vectors in Cartesian coordinates – in physics. We can use the unit vectors

$$\hat{x}, \quad \hat{y}, \quad \text{and} \quad \hat{z}$$

(or, if you prefer, $\hat{i}$, $\hat{j}$, and $\hat{k}$) to specify the three coordinate directions. These unit vectors can be used to write any other vector in this coordinate system, so for example the vector $\mathbf{A}$ can be written as

$$\mathbf{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z},$$

where A_x, A_y, and A_z and so on are the components of the vector along those directions. The unit vectors are called *complete* if they can be used to write every possible vector in this way.

In addition, these are *unit* vectors, which means that have a length of exactly one (with no dimensions). Mathematically we could express this using the dot product,

$$\hat{x} \cdot \hat{x} = 1, \quad \hat{y} \cdot \hat{y} = 1, \quad \hat{z} \cdot \hat{z} = 1.$$

We say that a vector is *normalized* if this is the case – that the dot product with itself is one.

Of course, these unit vectors are also *orthogonal* – they have 90° between them – which we could also write in terms of dot products as

$$\hat{x} \cdot \hat{y} = 0, \quad \hat{y} \cdot \hat{z} = 0, \quad \hat{z} \cdot \hat{x} = 0.$$

Taken together, we can use the term *orthonormal* for vectors that are both normalized and orthogonal.

The quantum state vectors $|+\rangle$ and $|-\rangle$ play a similar role as the unit vectors, but for spin-1/2 systems; because these states correspond to measuring spin up and spin down along the z axis, these kets form what we'll call the "S_z basis." And like $\hat{x}$, $\hat{y}$, and $\hat{z}$, they're complete, so that any general spin-1/2 state can be written down in terms of them:

$$|\psi\rangle = a|+\rangle + b|-\rangle. \tag{2.2}$$

The values a and b – which can be complex numbers, so be careful – in a loose sense[1] tell you how much of the general state $|\psi\rangle$ is made up of the spin up state (a) how much is made up of the spin down state (b).

Remember I introduced the bra $\langle\psi|$ above? Well, in the S_z basis the state could also be written as

$$\langle\psi| = a^*\langle+| + b^*\langle-|. \tag{2.3}$$

Those stars indicate the *complex conjugate*. It's important to realize that not only are the two representations of the state – the ket $|\psi\rangle$ and the bra $\langle\psi|$ – different (one obviously involves the complex conjugate of the other), but they're not even the same kind of mathematical object. That said, either provides a full representation of the same state.

Okay, so what about these funny names, *ket* and *bra*? They come about because of how we write the *inner product* – the analog to the dot

[1] Careful with taking this too far; we'll see exactly what a and b tell us soon.

product we used above for the unit vectors. We stick the bra first, then the ket, like so:

$$\langle \text{bra}|\text{ket}\rangle.$$

Can you see what the whole thing spells? I think this is supposed to be a joke, so feel free to laugh; we're stuck with the notation, though, which Paul Dirac invented in 1939, and which we now call *Dirac notation*.

With the inner product defined, we can now write out the normalization condition for our basis vectors,

$$\langle +|+\rangle = \langle -|-\rangle = 1, \tag{2.4}$$

as well as orthogonalization,

$$\langle +|-\rangle = \langle -|+\rangle = 0. \tag{2.5}$$

Actually, in quantum mechanics, it's not just the basis vectors that are normalized; *every* quantum state must be normalized:

$$\boxed{\langle \psi|\psi\rangle = 1.} \tag{2.6}$$

I know I keep saying this, but we'll see why this must the case later.

Example 2.1 – Normalization. Consider the state

$$|\psi\rangle = A\left(|+\rangle + 2i|-\rangle\right). \tag{2.7}$$

Is the state normalized? Check:

$$\begin{aligned} \langle \psi|\psi\rangle &= 1 \\ \Rightarrow A^*\left(\langle +| - 2i\langle -|\right)\ A\left(|+\rangle + 2i|-\rangle\right) &= 1 \end{aligned}$$

Expanding out the brackets and using the orthonormality conditions above gives

$$5|A|^2 = 1, \tag{2.8}$$

where $|A|^2 \equiv A^*A$ is the "complex square." Now, the complex square will always be a real number, which we can see easily if we write the general complex number A in polar form as

$$A = re^{i\theta},$$

where r (the *amplitude*) and θ (the *phase*) are both real numbers. Then

$$|A|^2 = A^*A = re^{-i\theta}re^{i\theta} = r^2.$$

That means the normalization condition above in equation (2.8) can't give us any information about the phase θ, only the amplitude. As it will turn out, that's okay – an overall phase in a quantum state doesn't mean anything physically, and we'll see why in due time – so by convention we take it to be zero, and set the constant to

$$A = \frac{1}{\sqrt{5}}.$$

With this value of A, the state is normalized.

Example 2.2 – Inner products. Consider another state given by

$$|\phi\rangle = \frac{3}{5}|+\rangle - \frac{4}{5}|-\rangle. \tag{2.9}$$

Notice that this state is already normalized (check it!). What is the inner product between state $|\psi\rangle$ and $|\phi\rangle$?

Using Dirac notation we can write each state in their S_z basis and "foil" it out to get

$$\begin{aligned}\langle\psi|\phi\rangle &= \left(\frac{1}{\sqrt{5}}\langle+| - \frac{2i}{\sqrt{5}}\langle-|\right)\left(\frac{3}{5}|+\rangle - \frac{4}{5}|-\rangle\right) \\ &= \frac{3}{5\sqrt{5}} + \frac{8i}{5\sqrt{5}}.\end{aligned}$$

Notice what happens if we reverse the order of the inner product:

$$\begin{aligned}\langle\phi|\psi\rangle &= \left(\frac{3}{5}\langle+| - \frac{4}{5}\langle-|\right)\left(\frac{1}{\sqrt{5}}|+\rangle + \frac{2i}{\sqrt{5}}|-\rangle\right) \\ &= \frac{3}{5\sqrt{5}} - \frac{8i}{5\sqrt{5}}.\end{aligned}$$

This is precisely the complex conjugate of the first inner product; in fact, in general for any two state vectors $|\alpha\rangle$ and $|\beta\rangle$ we have

$$\boxed{\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^*.} \tag{2.10}$$

2.2 Matrix Notation

Dirac notation is fun and expressive, but there's another way to write out a quantum state that can be a little more practical for calculations. Take a look at the form of some general state, which in Dirac notation is

$$|\psi\rangle = a|+\rangle + b|-\rangle. \tag{2.11}$$

Now, the kets $|+\rangle$ and $|-\rangle$ are the basis states as we've mentioned, and a and b are two complex numbers that tell you – again, loosely – how much of each basis state makes up the state $|\psi\rangle$. We can "pick out" these numbers with an inner product; for example,

$$\langle +|\psi\rangle = \langle +|\left(a|+\rangle + b|-\rangle\right) = a\langle +|+\rangle + b\langle +|-\rangle = a.$$

Notice how the bra $\langle +|$ acts past the numbers a and b to stick to the kets, and that to make the last equality we used orthonormality. Likewise, the inner product

$$\langle -|\psi\rangle = b.$$

As long as we know the basis we're in, any state can be specified by only those two numbers a and b. So let's use just them and write the state as a *column vector*,

$$|\psi\rangle \to \begin{pmatrix} a \\ b \end{pmatrix}. \tag{2.12}$$

I'm using an arrow ($\to$) rather than an equal sign since we're mixing notations here; the ket $|\psi\rangle$ is written in Dirac notation, and the column vector is what we'll call *matrix notation*. We could also write the column vector as

$$|\psi\rangle \to \begin{pmatrix} \langle +|\psi\rangle \\ \langle -|\psi\rangle \end{pmatrix}, \tag{2.13}$$

which mixes the two notations even further but makes it clear what basis the state is written in. It also makes it clear that the basis states themselves are just unit column vectors:

$$|+\rangle \to \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |-\rangle \to \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.14}$$

But what about the bra – how do we represent that? With a bit of thought, you might agree that they should be *row vectors*, so that the general state is written as

$$\langle\psi| \to \begin{pmatrix} a^* & b^* \end{pmatrix} \tag{2.15}$$

or

$$\langle\psi| \to \begin{pmatrix}\langle\psi|+\rangle & \langle\psi|-\rangle\end{pmatrix}. \tag{2.16}$$

We're really leveraging all this notation – the row vector elements are the complex conjugate of the numbers a and b, so in equation (2.16) I've swapped the order of the two states in the inner product.

Example 2.3 – Inner products, again. Let's go back to the previous example and recompute the inner product $\langle\psi|\phi\rangle$ but using matrix notation. Each state can now be written as

$$|\psi\rangle = \begin{pmatrix}1/\sqrt{5}\\ 2i/\sqrt{5}\end{pmatrix} \quad \text{and} \quad |\phi\rangle \to \begin{pmatrix}3/5\\ -4/5\end{pmatrix}.$$

Remember the bra is the row vector (with complex conjugate elements) and the ket the column, so the inner product works out perfectly:

$$\langle\psi|\phi\rangle \to \begin{pmatrix}1/\sqrt{5} & -2i/\sqrt{5}\end{pmatrix}\begin{pmatrix}3/5\\ 4/5\end{pmatrix} = \frac{3}{5\sqrt{5}} + \frac{8i}{5\sqrt{5}}.$$

This is of course the same result we got last time, but if you're like me you find it a little easier to multiply matrices than foil out in Dirac notation.

Problems

Problem 2.1 – Spin-1/2 States. Consider these different spin-1/2 states, given in Dirac notation by

$$\begin{aligned}|\psi_1\rangle &= 3|+\rangle + 4|-\rangle\\ |\psi_2\rangle &= |+\rangle + 2i|-\rangle\\ |\psi_3\rangle &= 3|+\rangle - e^{i\pi/3}|-\rangle\\ |\psi_4\rangle &= 3|+\rangle - 5i|-\rangle.\end{aligned}$$

(a) Normalize each state.

(b) Write each state in matrix notation.

(c) Write the bra of each ket; write the bras in matrix notation, too.

(d) Find a (normalized) ket that is orthogonal to each state.

(e) Compute the following inner products, using either Dirac notation or matrix notation: $\langle\psi_1|\psi_2\rangle$, $\langle\psi_3|\psi_4\rangle$, $\langle\psi_4|\psi_1\rangle$, and $\langle\psi_3|\psi_2\rangle$.

Problem 2.2 – A Three-State System. So far we've only looked at the fairly simple two-state spin-1/2 system. For this problem, consider a system which has *three* orthonormal basis states, given by

$$|a_1\rangle, |a_2\rangle, \text{ and } |a_3\rangle.$$

(a) What would these kets look like in matrix notation?

(b) Normalize the state

$$|\psi_1\rangle = |a_1\rangle - 2|a_2\rangle + 5|a_3\rangle.$$

(c) Take the inner product of $|\psi_1\rangle$ with the state

$$|\psi_2\rangle = i|a_1\rangle + 3|a_2\rangle - 2|a_3\rangle.$$

(you'll have to normalize that state first, of course).

Chapter 3

Amplitude and Probability

3.1 The S_x and S_y Spin States

In the previous chapter we talked about the two basis states – for what we're calling the S_z basis – and wrote them in Dirac and matrix notation as

$$|+\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |-\rangle \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.1}$$

But since these are basis states, *any* other state can be written in terms of them, including the states corresponding to measuring $S_x = +\hbar/2$ and $S_x = -\hbar/2$ that we discussed back in Experiment 2 in Section 1.3.2. I'll show you how to figure this out in Problem 3.6, and for now I'll just tell you the states:

$$|+\rangle_x = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \tag{3.2}$$

and

$$|-\rangle_x = \frac{1}{\sqrt{2}}|+\rangle - \frac{1}{\sqrt{2}}|-\rangle \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}. \tag{3.3}$$

Of course we can also rotate the Stern-Gerlach device so that we measure the spin along the y-direction, in which case we'd have the associated states $|+\rangle_y$ and $|-\rangle_y$. In the S_z basis, they can be written as

$$|+\rangle_y = \frac{1}{\sqrt{2}}|+\rangle + \frac{i}{\sqrt{2}}|-\rangle \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix} \tag{3.4}$$

and

$$|-\rangle_y = \frac{1}{\sqrt{2}}|+\rangle - \frac{i}{\sqrt{2}}|-\rangle \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}. \tag{3.5}$$

Now that we know how to write these states down, we can talk more about the inner product and its role in quantum mechanics.

3.2 Calculating Probability

Suppose you have a particle in the state $|\psi\rangle$, and you want to measure some quantity associated with the particle – maybe its spin, but it could be anything, so I'll call it A for now. In quantum mechanics we call this quantity an *observable*. Further suppose that there are a number of different possibilities that you might measure: a_1, a_2, a_3, and so on. Then the probability of measuring the particular value a_n is

$$\boxed{\mathscr{P}_{a_n} = |\langle a_n|\psi\rangle|^2.} \tag{3.6}$$

The bra in that equation corresponds to the state associated with measuring the value a_n; the ket would be $|a_n\rangle$.

This is one of the most important ideas in quantum mechanics, and is central to what's called the Born rule or the generalized statistical interpretation. The underlying idea – which we'll come back to at the end of this chapter – is that the theory of quantum mechanics can't predict exactly what value an experiment will produce (it can't tell you, when you measure A, if you will get a_1 or a_2 or something else) but only the probability of measuring each (so maybe 50% of the time you'll measure a_1, 25% of the time a_2, and so on).

Let's try applying this to our measurements of spin. If we want to measure the value of S_z, for example, the probability that we would get the value $S_z = +\hbar/2$ would be calculated by

$$\mathscr{P}_+ = |\langle +|\psi\rangle|^2,$$

and the probability of measuring $S_z = -\hbar/2$ is

$$\mathscr{P}_- = |\langle -|\psi\rangle|^2.$$

Of course, with only two possibilities, we'd have to have

$$\mathscr{P}_+ + \mathscr{P}_- = 1$$

so that we have 100% chance of measuring something.

One last thing: notice the role of the inner product, which in this context is called the *amplitude*, and that complex squaring the amplitude gives us the probability. Remember that the complex conjugate of an inner product flips the order, so the full complex square can be written

$$|\langle a_n|\psi\rangle|^2 = \left(\langle a_n|\psi\rangle\right)^* \langle a_n|\psi\rangle = \langle\psi|a_n\rangle\langle a_n|\psi\rangle. \tag{3.7}$$

Example 3.1 – Calculating probability. Suppose our particle is in the state $|+\rangle$. In this state, of course, there's a 100% chance of measuring $S_z = +\hbar/2$, as we can see with the calculation

$$\mathscr{P}_+ = |\langle +|+\rangle|^2 = 1.$$

But what's the probability of measuring $S_x = +\hbar/2$? This is easier to calculate using matrix notation:

$$\mathscr{P}_{+x} = |_x\langle +|+\rangle|^2 \to \left|\begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}\right|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}.$$

So there's a 50% chance of measuring spin up along the x-direction.

We went over this already back in the Stern-Gerlach Experiment 1 (Section 1.3.1) and Experiment 2 (Section 1.3.2) – this is the theory that explains the results we discussed then. So we might as well go over Experiment 3, too (Section 1.3.3), in which we had a beam of particles in the state $|+\rangle_x$ entering a Stern-Gerlach device aligned to measure S_z. Question: how many particles will we measure to have $S_z = +\hbar/2$? Well, the probability of measuring spin up along z is

$$\mathscr{P}_+ = |\langle +|+\rangle_x|^2 \to \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}\right|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}.$$

So 50% of the particles should have $S_z = +\hbar/2$ (and presumably the other 50% are spin down).

3.3 Indeterminant States

Let's return to an issue that I've been trying to avoid. As usual, pretend we have a particle in some general state $|\psi\rangle$, and suppose we want to measure the observable S_z. We know from our Stern-Gerlach discussions that the state *changes* once a measurement is made – we say the state *collapses* to the state that corresponds to the measurement we made. For example, if a particle is in state $|\psi\rangle$ and we measure $S_z = +\hbar/2$, the state has collapsed to the new state $|+\rangle$. This, though, is not the issue I've been dancing around.

Instead, consider this question: what is the value of S_z *just before* we make the measurement? I know what it is after measurement – it's $+\hbar/2$ – and I can check by measuring S_z again (and I'll get the same value as

long as the state hasn't been disturbed in some way). But what's S_z – the component of spin angular momentum along the z-direction – *before* that first measurement?

This is a more difficult question to answer than you might think since our theory doesn't quite provide an answer for us – the fact that the state collapses upon measurement means that we don't really know the value of S_z before the collapse. There are three possibilities we must consider:[1]

1. The obvious possibility is that the spin was $S_z = +\hbar/2$ just before measurement, too – that's what we measure it to be, after all. This is called the *realist* position. There's just one problem, though – if the particle has $S_z = +\hbar/2$ all along, why couldn't quantum mechanics tell us that? This suggests that it's an *incomplete theory* – we're missing something that would allow us to calculate directly the answer to this question.

2. Another option is a little more drastic – maybe the particle did not have a value for S_z. Not that it's zero or we don't know the value – but that it does not have a definite value for its spin angular momentum. This is called *indeterminancy* – the state is indeterminant with respect to its value of S_z – and this interpretation of quantum mechanics is called the *Copenhagen interpretation*.

3. I guess we could also just refuse to answer – the theory doesn't allow us to answer, so maybe it's not a good question to ask. This is called the *agnostic* position.

In 1964 John Bell showed that it actually makes an observable difference whether a particle has a value of an observable (even if it's unknown) or if it's indeterminant. Although I won't get into the details of Bell's work here,[2] the conclusion is pretty remarkable: it's the Copenhagen interpretation that is the correct one. Actually, I should be a little more careful, since there are extensions of quantum mechanics (called nonlocal hidden variable theories) that can maintain the realist position, as well as interpretations (like the popular many-worlds interpretation) that fall outside of the simple picture I've gone over here. But those are also outside the scope of this book, so going forward we'll adopt the Copenhagen interpretation in which observables might not have precise values before measurement.

[1] From Griffiths' *Introduction to Quantum Mechanics*, section 1.2
[2] See Griffiths *Introduction to Quantum Mechanics* section 12.2.

Problems

Problem 3.1 – Orthogonality. Show that the states $|+\rangle_x$ and $|-\rangle_x$ are orthogonal. Do the same for the states $|+\rangle_y$ and $|-\rangle_y$.

Problem 3.2 – Measurements. Using a Stern-Gerlach device rotated at some angle, you're able to prepare a beam of particles all in the state (we'll see later on how you can do this)

$$|\psi\rangle = 3|+\rangle + 5i|-\rangle.$$

(a) What is the normalized state?

(b) If you were to measure S_z, what are the possible results and their probabilities?

(c) Suppose you measured S_z on the beam of particles, and some of them were found to be $-\hbar/2$. If you then made a measurement of S_y on just those particles, what are the possible results and their probabilities?

Problem 3.3 – S_z and S_x Measurements. Consider the two normalized states

$$|\psi_1\rangle = \frac{2}{\sqrt{13}}|+\rangle + \frac{3i}{\sqrt{13}}|-\rangle \quad \text{and} \quad |\psi_2\rangle = \frac{2}{\sqrt{13}}|+\rangle_x + \frac{3i}{\sqrt{13}}|-\rangle_x.$$

(a) If you measure the observable S_z on each state, what are the possible results you would measure, and with what probability would you measure each?

(b) If you measure the observable S_x on each state, what are the possible results you would measure, and with what probability would you measure each?

Problem 3.4 – An Overall Phase.

(a) First, consider this normalized state:

$$|\psi_1\rangle = \frac{4}{5}|+\rangle + \frac{3i}{5}|-\rangle.$$

What is the probability of measuring $S_y = +\hbar/2$ for a particle in this state?

(b) Next, we can change the state by introducing an *overall phase factor*,

$$|\psi_2\rangle = -(\frac{4}{5}|+\rangle + \frac{3i}{5}|-\rangle) = -\frac{4}{5}|+\rangle - \frac{3i}{5}|-\rangle.$$

What is the probability of measuring $S_y = +\hbar/2$ for a particle in this state? What affect does the overall phase factor have on the measurement?

(c) We can also change the state by introducing a *relative phase factor*,

$$|\psi_3\rangle = \frac{4}{5}|+\rangle - \frac{3i}{5}|-\rangle.$$

What is the probability of measuring $S_y = +\hbar/2$ for a particle in this state? What affect does the relative phase factor have on the measurement?

Problem 3.5 – An Overall Phase Part 2.

(a) Show that you can write the overall phase from Problem 3.4 (b) as

$$|\psi_2\rangle = e^{i\delta}\left(\frac{4}{5}|+\rangle + \frac{3i}{5}|-\rangle\right).$$

What is the value of δ?

(b) Show that an overall phase for *any* measurement on a general state $|\psi\rangle$ has no affect on the probability. This is a fact we'll exploit every so often – that an overall phase has no affect on a physical measurement.

Problem 3.6 – The S_x States. You might be wondering how I was able to write down the states for $|+\rangle_x$ and $|-\rangle_x$ back at the start of this chapter. This problem will walk you through the details – it comes from analysing the Stern-Gerlach experiments of Chapter 1. From Experiment 2 (or Experiment 3), we can write down a bunch of probabilities based on the results:

$$|\langle +|+\rangle_x|^2 = \tfrac{1}{2} \quad |\langle +|-\rangle_x|^2 = \tfrac{1}{2}$$
$$|\langle -|+\rangle_x|^2 = \tfrac{1}{2} \quad |\langle -|-\rangle_x|^2 = \tfrac{1}{2}.$$

(a) Write the unknown kets $|+\rangle_x$ and $|-\rangle_x$ as

$$|+\rangle_x = a|+\rangle + be^{i\alpha}|-\rangle \quad \text{and} \quad |-\rangle_x = c|+\rangle + de^{i\beta}|-\rangle,$$

where a, b, c, and d as well as α and β are all *real* numbers – we're leveraging the fact the only relative phases are important as discussed in Problem 3.5. Use these kets and the probability results above to find the unknowns a, b, c, and d.

(b) For the numbers α and β we need to use a bit more information about our states. We know, for example, they should be normalized (which they already are) and that they should be orthogonal:

$$ {}_x\langle -|+\rangle_x = 0. $$

Use this property to show that we must have

$$ e^{i\alpha} = -e^{i\beta}. $$

(c) That's as far as we can go – we have some freedom to choose the phases α and β as long as this equation is satisfied (this corresponds exactly to the fact that the z and x axes have to be perpendicular – but there's an infinite number of directions the x axis can have and still be perpendicular). By convention, take $\alpha = 0$; what value of β is then necessary?

(d) Use all your now-found unknowns to write out the two states $|+\rangle_x$ and $|-\rangle_x$, and confirm that they match equations (3.2) and (3.3).

What about the S_y states? Go ahead and try if you're interested; all the probabilities will be the same, the only difference is that $|+\rangle_y$ and $|-\rangle_y$ must also satisfy the probability relations between an x Stern-Gerlach device and a y one, and we have to choose $\alpha = \pi/2$ for that.

Problem 3.7 – A Three-State System. Remember our three-state system from Problem 2.2? We had three orthonormal basis states, given by

$$ |a_1\rangle, |a_2\rangle, \text{ and } |a_3\rangle. $$

Here, each state corresponds to a measurement of an observable A.

(a) Suppose the system is prepared in the state

$$ |\psi\rangle = |a_1\rangle - 2|a_2\rangle + 5|a_3\rangle. $$

You then measure the observable A on this state; what are the possible results and probabilities of each?

(b) What is the probability of measuring the system to be in the state

$$ |\phi\rangle = (1+i)|a_1\rangle + |a_2\rangle + |a_3\rangle? $$

Chapter 4

Observables and Operators

4.1 Operators and Measurement

We've already seen lots of examples of *observables* – the spin components S_x, S_y, and S_z are all examples of things that we can observe, as are things like position and momentum. In the language of quantum mechanics:

> A physical observable A is represented by an operator $\hat{A}$.

Wait, what exactly is an operator? At the risk of being too simplistic, it's something that *operates* on a quantum state, producing a new state. Mathematically we can write this operation like

$$\hat{A}|\psi\rangle = |\phi\rangle.^{1} \tag{4.1}$$

Actually, although in general the operator $\hat{A}$ produces a new state, sometimes the state it operates on is special and *doesn't* change:

$$\hat{A}|\psi\rangle = a|\psi\rangle. \tag{4.2}$$

Instead of making a new state, it gives back the same state, but multiplied by a number a. If you're familiar with linear algebra, you might recognize equation (4.2) as an *eigenvalue equation*, with a playing the role of the eigenvalue and the ket $|\psi\rangle$ being the eigenvector. The eigenvalue is very important:

> The only possible result of a measurement of an observable A is one of the eigenvalues a_n of the corresponding operator $\hat{A}$.

[1] Although the hat on the operator is standard, many textbooks drop it to reduce clutter in notation. I think it's important to be able to easily distinguish an observable and an operator at a glance, especially for newcomers.

4.2 The Operator $\hat{S}_z$

That's a lot to take in on first reading, so let's do a full example. We'll return to the observable S_z, which must have a corresponding operator $\hat{S}_z$. We'll figure out how we can write this in a moment, but you might have realized we already know the eigenvalues of this operator: they must be $\hbar/2$ and $-\hbar/2$, since we know measurements of S_z always return one of those two numbers.

That means that the eigenvalue equations – two of them for two eigenvalues – could be written as

$$\hat{S}_z|+\rangle = \frac{\hbar}{2}|+\rangle \quad \text{and} \quad \hat{S}_z|-\rangle = -\frac{\hbar}{2}|-\rangle. \tag{4.3}$$

We can get a sense of what the operator $\hat{S}_z$ is by thinking of this in matrix notation – the states $|+\rangle$ and $|-\rangle$ are column vectors, meaning $\hat{S}_z$ must be represented as a 2×2 matrix (nothing else, when multiplied by a column vector, gives another column vector). I don't know what the elements of the matrix are, so I'll write it out for now as

$$\hat{S}_z \to \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

To find the unknown elements a, b, c, and d (which could be complex), we can go back to the eigenvalue equations, this time in matrix notation. For the eigenvalue $\hbar/2$, we have

$$\hat{S}_z|+\rangle = \frac{\hbar}{2}|+\rangle \to \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{4.4}$$

Multiplying out the left side gives

$$\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} \hbar/2 \\ 0 \end{pmatrix},$$

so we know that $a = \hbar/2$ and $c = 0$. The other eigenvalue equation, for $-\hbar/2$, similarly gives $b = 0$ and $d = -\hbar/2$. The operator $\hat{S}_z$, in matrix notation, is

$$\hat{S}_z \to \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.5}$$

There's a few things here we should notice – in particular that $\hat{S}_z$ is a *diagonal* matrix, with the eigenvalues as the diagonal elements. Also note that the eigenvectors (the states $|+\rangle$ and $|-\rangle$) are unit vectors. Both of these things arise because of the basis were using – it's the S_z basis! In other words, in its own basis – where the basis states are the eigenstates of the basis operator – an operator will be diagonal with the eigenvalues along that diagonal.

4.3 Matrix Representation

In fact, any operator can be represented as a matrix; sticking with the S_z basis, we can write a general operator $\hat{A}$ as a 2×2 matrix

$$\hat{A} \to \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{4.6}$$

Now for something a little new. Consider the calculation $\langle + | \hat{A} | + \rangle$. What exactly does this mean? First we apply the operator $\hat{A}$ to the ket $|+\rangle$, getting a new state, and then apply the bra $\langle +|$ to that new state:

$$\langle + | \left(\hat{A} | + \rangle \right) \to \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = a.$$

So the ultimate result is just a number – a in this case. Similarly, you can show that $\langle + | \hat{A} | - \rangle = b$, and so on. That means we could be able to write our operator in matrix notation as

$$\hat{A} \to \begin{pmatrix} \langle + | \hat{A} | + \rangle & \langle + | \hat{A} | - \rangle \\ \langle - | \hat{A} | + \rangle & \langle - | \hat{A} | - \rangle \end{pmatrix}. \tag{4.7}$$

This is just notation, but useful since we can see explicitly what the basis is.

It's easy enough to generalize this to any system; the matrix representation might have a different size than 2×2 (it might even be infinite), but the idea is the same. In general we can write

$$\hat{A} \to \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ a_{31} & a_{32} & a_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{4.8}$$

where $a_{ij} = \langle i | \hat{A} | j \rangle$ and $|i\rangle$ and $|j\rangle$ are the basis states. In this context we'll often call the a_{ij}s matrix elements.

4.4 The Operators $\hat{S}_x$ and $\hat{S}_y$

The matrix representation of the other two spin component operators are given by

$$\hat{S}_x \to \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{4.9}$$

and

$$\hat{S}_y \to \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{4.10}$$

These are both written in the S_z basis (notice that they're not diagonal). If you'd like to know where these came from, feel free to work through Problems 4.1 and 4.2. What I'd rather do right now is ask the question: *if we were to measure, say, S_y on some particular state, what are the possible results we could get?* We already know the answer in this case, but a lot of quantum mechanics boils down to figuring out the answer to this question for a variety of different observables and corresponding operators, so it's worth going through the process in detail.

The answer, as mentioned above, is given by the eigenvalues of the operator. You probably already know how to find eigenvalues from your linear algebra course, but let's do it together nice and slow. First, the eigenvalue equation is

$$\hat{S}_y|\lambda\rangle = \lambda|\lambda\rangle, \tag{4.11}$$

where λ does double duty here, standing for the unknown eigenvalue as well as the corresponding eigenstate. This might seem confusing at first, but we'll get used to it.

We find the eigenvalues by solving the characteristic (or secular) equation , which looks like

$$\det|\mathbf{S}_y - \lambda\mathbf{I}| = 0. \tag{4.12}$$

The bolded letter $\mathbf{S}_y$ denotes the matrix representation of the operator, and $\mathbf{I}$ is the identity matrix. Filling in the matrices gives

$$\left|\begin{pmatrix} 0 & -i\hbar/2 \\ i\hbar/2 & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right| = 0,$$

or, after simplifying,

$$\begin{vmatrix} -\lambda & -i\hbar/2 \\ i\hbar/2 & -\lambda \end{vmatrix} = 0.$$

Work out the determinant to get

$$\lambda^2 - \left(\frac{\hbar}{2}\right)^2 = 0$$

so that, finally, we can see that the two eigenvalues – as expected! – are $\lambda_1 = \hbar/2$ and $\lambda_2 = -\hbar/2$. These two values are the only possible results we would get if we measured S_y on any spin state.

What about the eigenstates? To find these, we go back to the original eigenvalue equation (4.11) and plug in each eigenvalue in turn. For $\lambda = \hbar/2$, we have

$$\hat{S}_y|\lambda\rangle = \frac{\hbar}{2}|\lambda\rangle.$$

Writing the unknown state $|\lambda\rangle$ in matrix notation,

$$|\lambda\rangle \to \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \tag{4.13}$$

our job is to find the value of α and β that satisfy the eigenvalue equation. Here we go:

$$\hat{S}_y|\lambda\rangle = \frac{\hbar}{2}|\lambda\rangle \to \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}\alpha\\ \beta\end{pmatrix},$$

so

$$\begin{pmatrix}-i\beta\\ i\alpha\end{pmatrix} = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}.$$

This is really two equations for α and β, but they're the same equation – we don't have enough information to get both values right now. But we can at least solve for β in terms of α and get

$$\beta = i\alpha,$$

making the eigenstate so far

$$|\lambda\rangle = |+\rangle_y \to \alpha\begin{pmatrix}1\\ i\end{pmatrix}. \tag{4.14}$$

How, then, do we get the last unknown α? Well, every state must be normalized, so

$${}_y\langle +|+\rangle_y = 1. \tag{4.15}$$

In matrix notation, this becomes

$$|\alpha|^2\begin{pmatrix}1 & -i\end{pmatrix}\begin{pmatrix}1\\ i\end{pmatrix} = |\alpha|^2(2) = 1,$$

so we'll take

$$\alpha = \frac{1}{\sqrt{2}}$$

and therefore

$$|+\rangle_y = \frac{1}{\sqrt{2}}|+\rangle + \frac{i}{\sqrt{2}}|-\rangle \to \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}. \tag{4.16}$$

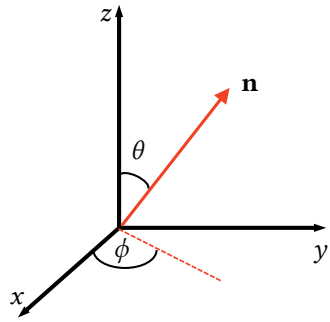


Figure 4.1: A random direction for the alignment of the Stern-Gerlach device.

Feel free to go through the same process to get the spin down eigenstate; it turns out to be

$$|-\rangle_y = \frac{1}{\sqrt{2}}|+\rangle - \frac{i}{\sqrt{2}}|-\rangle \rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}. \tag{4.17}$$

Problem 4.1 will get you to give all this a try for S_x.

4.5 Spin in a General Direction, $\hat{S}_n$

At this point I just want to give you some more operators and figure out their eigenvalues and eigenstates, partly so we can get the practice and partly so we can use them again later. First up, suppose we rotate our usual Stern-Gerlach device to some random orientation so that the magnetic field points in the direction of $\mathbf{n}$ as shown in Figure 4.1. The angles θ (the polar angle) and ϕ (the azimuthal angle) are the usual spherical coordinates, and we'll make $\mathbf{n}$ a unit vector so it has a length of one. That means in terms of the Cartesian unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, we can write

$$\mathbf{n} = \sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}}. \tag{4.18}$$

Suppose we measure the spin of a particle along this direction; let's call the observable S_n, where

$$S_n = \mathbf{S}\cdot\mathbf{n} = S_x\sin\theta\cos\phi + S_y\sin\theta\sin\phi + S_z\cos\theta. \tag{4.19}$$

The operator associated with this observable is, of course, the exact same except with hats on the operators, so we have[2]

$$\hat{S}_n = \hat{\mathbf{S}} \cdot \mathbf{n} = \hat{S}_x \sin\theta\cos\phi + \hat{S}_y \sin\theta\sin\phi + \hat{S}_z \cos\theta. \tag{4.20}$$

To write this operator in matrix notation, we just fill in the matrices for $\hat{S}_x$, $\hat{S}_y$, and $\hat{S}_z$ and add them together to get

$$\hat{S}_n \to \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\, e^{-i\phi} \\ \sin\theta\, e^{i\phi} & -\cos\theta \end{pmatrix} \tag{4.21}$$

(see Problem 4.4).

This is a new operator we've never seen before, and anytime we're faced with a new operator, we should always find its eigenvalues and eigenstates – remember, the eigenvalues of the operator answer the question of what we get when we measure the observable, so they're pretty important, and we'll see that the eigenstates play just as important a role. But wait – we *already* know the eigenvalues! They *have* to be $\hbar/2$ and $-\hbar/2$, just like $\hat{S}_x$ or any other component. After all, changing the direction of the magnetic field by rotating the device is the exact same operation as changing our coordinate system, and that shouldn't affect the actual results we get.

So, fine, we know the eigenvalues; what about the eigenstates? You need more practice than I do, so try on your own (that's Problem 4.4 again), and I'll give you just the answers for now. The eigenstates are (with notation that should be clear by now; the ket $|+\rangle_n$ is the state that goes with $+\hbar/2$ and similarly for the eigenvalue $-\hbar/2$)

$$|+\rangle_n = \cos(\theta/2)\,|+\rangle + \sin(\theta/2)\, e^{i\phi}\,|-\rangle \tag{4.22}$$

and

$$|-\rangle_n = \sin(\theta/2)\,|+\rangle - \cos(\theta/2)\, e^{i\phi}\,|-\rangle. \tag{4.23}$$

Remember, all this is explicitly in the S_z basis.

By the way, if you were wondering how we make some "general spin state" $|\psi\rangle$ with some specific set of coefficients, this is how: orient the Stern-Gerlach device at just the correct angles θ and ϕ that give you those coefficients. Problem 4.5 will ask you to do this.

[2]Okay, we have to be a little careful now distinguishing unit vectors from operators, both of which have hats. Things should be clear from context.

4.6 The $\hat{S}^2$ Operator

All of the spin operators we've seen so far have been *components* of the spin angular momentum *vector*. We know that we can't simultaneously know all three components, but it turns out we can measure the total magnitude of the spin; in fact, we usually work with the square,

$$S^2 = S_x^2 + S_y^2 + S_z^2. \tag{4.24}$$

At the risk of repeating things, the operator $\hat{S}^2$ is of course

$$\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2, \tag{4.25}$$

and we can determine what it looks like in the S_z basis using matrix notation. Substituting from equations (4.5), (4.9), and (4.10), we see that

$$\hat{S}^2 \rightarrow \left[\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\right]^2 + \left[\frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\right]^2 + \left[\frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\right]^2 \tag{4.26}$$

or

$$\hat{S}^2 \rightarrow \frac{3}{4}\hbar^2\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \tag{4.27}$$

But you might recognize this matrix as the identity matrix – and it's diagonal, which means we can just read off the eigenvalues: there's two but they're the same, $3\hbar^2/4$. That means that *every* measurement of S^2 (or just the magnitude S) will return $3\hbar^2/4$ (or $\sqrt{3}\hbar/2$). The eigenstates are simple, too, given the diagonal matrix – they're the basis states $|+\rangle$ and $|-\rangle$.

4.7 The Projection and Identity Operators

We've seen that one way to combine two quantum states is the *inner product*, which is related to probability, but there's also another way to combine two states called an *outer product*, which takes the form

$$|\phi\rangle\langle\psi|.$$

For example, consider the so-called *projection operators*, formed from the outer product of a basis state. One possibility is written as

$$\hat{P}_+ = |+\rangle\langle+|, \tag{4.28}$$

with the other (in our two basis-state system) as

$$\hat{P}_- = |-\rangle\langle-|. \tag{4.29}$$

Notice that these really are operators: the bra will act on another state, leaving some number times the ket.

Let's see what $\hat{P}_+$ does to a general state $|\psi\rangle = a|+\rangle + b|-\rangle$:

$$\hat{P}_+|\psi\rangle = \Big(|+\rangle\langle+|\Big)|\psi\rangle = |+\rangle\langle+|\Big(a|+\rangle + b|-\rangle\Big) = a|+\rangle.$$

We can see why it's called a projection operator: applying it to a state gives us a new state that is how much "spin up" the original state was times the spin up state itself. Similarly, of course, for $\hat{P}_-$:

$$\hat{P}_-|\psi\rangle = b|-\rangle.$$

What are the matrix representations of $\hat{P}_\pm$? In the S_z basis,

$$\hat{P}_+ \to \begin{pmatrix} \langle+|\hat{P}_+|+\rangle & \langle+|\hat{P}_+|-\rangle \\ \langle-|\hat{P}_+|+\rangle & \langle-|\hat{P}_+|-\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{4.30}$$

and

$$\hat{P}_- \to \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.31}$$

Here's another operator while we're at it: I'll define the *identity operator* as

$$\hat{1} = \hat{P}_+ + \hat{P}_-. \tag{4.32}$$

Take a look at the effect it has on a state:

$$\hat{1}|\psi\rangle = (\hat{P}_+ + \hat{P}_-)|\psi\rangle = \hat{P}_+|\psi\rangle + \hat{P}_-|\psi\rangle = a|+\rangle + b|-\rangle.$$

But this is exactly the state $|\psi\rangle$ that we started with – so there's no change at all.

It's even clearer if we work out the matrix representation; I'll let you check, but of course it's the identity matrix,

$$\hat{1} \to \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.33}$$

4.8 Hermitian Operators

We've just gone through lots of different operators, so it's time to circle back to the start of this chapter. We said there that an operator acts on a state and produces a new state:

$$\hat{A}|\psi\rangle = |\phi\rangle.$$

How does the operator act on a *bra*? In general, it produces a *different* state,

$$\langle\psi|\hat{A} = \langle\xi|. \tag{4.34}$$

Notice that the operator sits to the right of the bra and acts to the left, and that the state that it produces when acting on $\langle\psi|$ is *not* the same state as the operator acting on the *ket*; that is, $|\phi\rangle$ and $|\xi\rangle$ are different quantum states.

However, if we instead act on the bra $\langle\psi|$ with a different operator, $\hat{A}^\dagger$ – called the Hermitian adjoint of $\hat{A}$ – then we *do* get the state $\langle\phi|$:

$$\langle\psi|\hat{A}^\dagger = \langle\phi|. \tag{4.35}$$

Wait, what exactly is a Hermitian adjoint? To see, imagine taking the inner product of the state $|\phi\rangle = \hat{A}|\psi\rangle$ with some other state $|\beta\rangle$,

$$\langle\phi|\beta\rangle,$$

and of course we can flip the order by taking the complex conjugate,

$$\langle\phi|\beta\rangle = \langle\beta|\phi\rangle^*.$$

But $\langle\phi| = \langle\psi|\hat{A}^\dagger$ and $|\phi\rangle = \hat{A}|\psi\rangle$, so we can write

$$\langle\psi|\hat{A}^\dagger|\beta\rangle = \langle\beta|\hat{A}|\psi\rangle^*.$$

Remember, these operations are called matrix elements, so this tells us that the matrix representation of $\hat{A}^\dagger$ is the *transpose* and complex conjugate of $\hat{A}$.

One last thing, but it's really important: sometimes it turns out that $\hat{A}^\dagger = \hat{A}$ (or, in matrix notation, the operator is equal to its transpose conjugate). In this case we say that the operator $\hat{A}$ is *Hermitian*. In fact,

Every physical observable is represented by a Hermitian operator.

That's important, since you can prove that a Hermitian operator will always have *real* eigenvalues and that the eigenvectors will always be complete.

4.9 The Ladder Operators

So far every operator we've talked about has been Hermitian – go ahead and check. As a last example for the chapter, and one we'll make use of later, consider this mix of the operators $\hat{S}_x$ and $\hat{S}_y$:

$$\hat{S}_+ \equiv \hat{S}_x + i\hat{S}_y \quad \text{and} \quad \hat{S}_- \equiv \hat{S}_x - i\hat{S}_y. \tag{4.36}$$

These are called "ladder operators," with $\hat{S}_+$ being the *raising* operator and $\hat{S}_-$ the *lowering* operator.

What do these operators do? It's easiest to see in matrix notation, which is straightforward to figure out – just add the spin matrices together in the appropriate way. The raising operator is

$$\hat{S}_+ \to \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \tag{4.37}$$

and the lowering operator is

$$\hat{S}_- \to \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{4.38}$$

I think you can see at a glance that these matrices are not Hermitian – try taking their transpose. But while we're looking at them, note that $\hat{S}_+^\dagger = \hat{S}_-$; they're *Hermitian conjugates*.

Let's see what the raising operator does to our basis states, starting with spin up,

$$\hat{S}_+|+\rangle \to \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{0},$$

and then spin down,

$$\hat{S}_+|-\rangle \to \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \hbar \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \hbar|+\rangle.$$

What's going on here? The raising operator "raises" the spin down state to become spin up (times a constant), but, since there's no higher spin state than spin up, when $\hat{S}_+$ acts on that, it just gives zero.

Can you guess how the lowering operator will act? It should "lower" the spin up state, and give us zero for the spin down state, and it does:

$$\hat{S}_-|+\rangle \to \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \hbar|-\rangle,$$

and

$$\hat{S}_-|-\rangle \to \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \mathbf{0}.$$

Problems

Problem 4.1 – Matrix Representation of $\hat{S}_x$.

Our goal in this question is to find the matrix representation of the operator $\hat{S}_x$. Luckily the technique we'll use is completely general, so you can apply it to any other operator. First, write the operator as a matrix in the basis you're using – in our case, the S_z basis as usual – using the notation of equation (4.7),

$$\hat{S}_x \to \begin{pmatrix} \langle +|\hat{S}_x|+\rangle & \langle +|\hat{S}_x|-\rangle \\ \langle -|\hat{S}_x|+\rangle & \langle -|\hat{S}_x|-\rangle \end{pmatrix}.$$

Now we just have a lot of work to go through, figuring out each matrix element. Since we know the operation of $\hat{S}_x$ on its eigenstates,

$$\hat{S}_x|+\rangle_x = \frac{\hbar}{2}|+\rangle_x \quad \text{and} \quad \hat{S}_x|-\rangle_x = -\frac{\hbar}{2}|-\rangle_x$$

we should first convert the states $|+\rangle$ and $|-\rangle$ to the S_x basis and use that. Go ahead and give it a try – calculate all four matrix elements – and compare with equation (4.9).

Problem 4.2 – Matrix Representation of $\hat{S}_y$.

Following the procedure outlined in the question above, find the matrix represesntation of $\hat{S}_y$.

Problem 4.3 – Eigenvalues of $\hat{S}_x$.

In Section 4.4 we went through the process of finding the eigenvalues and eigenvectors of $\hat{S}_y$. Now it's your turn: find the eigenvalues of eigenvectors of $\hat{S}_x$.

Problem 4.4 – The $\hat{S}_n$ Operator.

(a) Given the form of the operator $\hat{S}_n$ in equation (4.20), show that the matrix formulation is as in equation (4.21).

(b) Using the matrix representation, find the eigenvalues and eigenstates of $\hat{S}_n$.

Problem 4.5 – Creating a Specific State. Suppose you're an experimental physicist tasked with creating a beam of spin-1/2 particles all in the state

$$|\psi\rangle = \frac{\sqrt{3}}{2}|+\rangle + \frac{i}{2}|-\rangle.$$

You take your trusty Stern-Gerlach device and set it up so that particles go from an oven, through collimaters, and then into the S-G device. You

rotate it by angles θ and ϕ and direct particles coming from the "spin-up" output to where you need them. What angles do you set so that all particles in this beam have the correct state?

Problem 4.6 – Spin-1.

Until now, we've been playing with spin-1/2 particles – electrons, protons, neutrons, and, apparently, silver atoms. However, our lab has just been gifted a new set of photons to play with, and photons are particles that have a spin of one. Sure enough, when we send them through a Stern-Gerlach type experiment to measure the photon's spin component along the z-direction, we get three possible results rather than two:

- $S_z = +\hbar$
- $S_z = 0$
- $S_z = -\hbar$.

This is the hallmark of a spin-1 system, and our goal in this assignment is to fully explore it.

(a) To start, we need to pick a basis, and given our experience with the S_z basis in spin-1/2, we should use that here, too. Write down the following three things: (i) the three basis states (call them $|1\rangle$, $|0\rangle$, and $|-1\rangle$, corresponding to measuring $+\hbar$, 0, and $-\hbar$ respectively), as kets and in matrix notation; (ii) the eigenvalue equations (all three of them since we have three eigenvalues) for the operator $\hat{S}_z$; and (iii) the matrix representation of the operator $\hat{S}_z$.

(b) Recall the raising and lowering operators we defined earlier:

$$\hat{S}_+ \equiv \hat{S}_x + i\hat{S}_y \quad \text{and} \quad \hat{S}_- \equiv \hat{S}_x - i\hat{S}_y.$$

After painstaking theoretical work, you finally figure out what they do when operating on our basis states:[3]

$$\begin{aligned} \hat{S}_+|1\rangle &= 0 & \hat{S}_-|1\rangle &= \sqrt{2}\hbar|0\rangle \\ \hat{S}_+|0\rangle &= \sqrt{2}\hbar|1\rangle & \hat{S}_-|0\rangle &= \sqrt{2}\hbar|-1\rangle \\ \hat{S}_+|-1\rangle &= \sqrt{2}\hbar|0\rangle & \hat{S}_-|-1\rangle &= 0. \end{aligned}$$

Use this to write out the matrix representation of $\hat{S}_+$ and $\hat{S}_-$, and then use their definition above to find the matrix representation of $\hat{S}_x$ and $\hat{S}_y$ (that was our ultimate goal here).

[3] Although it's obvious that they'll raise and lower the states, the $\sqrt{2}$ factor is a little harder; come see me if you're curious.

(c) Now that we have $\hat{S}_x$ and $\hat{S}_y$, we need to find their eigenvalues and eigenvectors. Do so, and write out the eigenstates in ket form and in matrix form.

(d) We now have a complete theory of spin-1, which means we can finally so some interesting experiments with those photons. Suppose you arrange your apparatus to prepare the photons in the state

$$|\psi\rangle = |1\rangle - 3|0\rangle + 2i|-1\rangle.$$

i. You prepare 1000 photons this way, and measure their spin component S_z. What values do you measure, and how many photons of the 1000 yield each result?

ii. You collect all the photons that had mesaurements of $S_z = -\hbar$. You then measure S_x on those; what do you get, and what is the number of photons for each possible result?

iii. If, on the original set of 1000 photons, you had instead measured S_y, what would have been the average of all results?

Chapter 5

Measurement and Uncertainty

5.1 Expectation Values

We've already discussed how quantum mechanics can't give us a prediction of a particular experiment – at best it can only give us probabilities of each possible outcome measured on a state $|\psi\rangle$:

$$\mathscr{P}_{a_n} = |\langle a_n|\psi\rangle|^2,$$

where a_n are the possible results. That's not very useful – what *else* can we extract from the theory?

Well, we can run the experiment many times and look at statistics of results. But we have to be careful: experiments should measure the same observable A each time, and, most importantly, the system must be prepared *identically* each time. Since measuring in quantum mechanics changes the state, we either have to have a way to revert back to the original state $|\psi\rangle$ or prepare a large number of such states and make the measurement on all of them simultaneously. This large number of identical states is called an *ensemble*.

For example, suppose we have an ensemble of $N = 100$ particles, all in the same state $|\psi\rangle$. At $t = 0$, we measure the spin component S_z on each (I guess we'll need 100 physics students and 100 Stern-Gerlach devices to do the job), and get

- N_+ particles that were measured to have $S_z = +\hbar/2$ and are now in state $|+\rangle$, and
- N_- particles that were measured to have $S_z = -\hbar/2$ and are now in state $|-\rangle$.

Remember, this doesn't need to be 50 and 50 – the state $|\psi\rangle$ could be *any* state – but we do need N_+ and N_- to add up to 100.

The mean or average of all results is

$$\langle S_z\rangle = \frac{\left(+\frac{\hbar}{2}\right)N_+ + \left(-\frac{\hbar}{2}\right)N_-}{N}, \tag{5.1}$$

or

$$\langle S_z\rangle = \left(+\frac{\hbar}{2}\right)\frac{N_+}{N} + \left(-\frac{\hbar}{2}\right)\frac{N_-}{N}.$$

But N_+/N is just the probability $\mathscr{P}_+$ of measuring $+\hbar/2$ (at least in the limit $N \to \infty$), and similarly for N_-/N, so we can write this as

$$\langle S_z\rangle = \left(+\frac{\hbar}{2}\right)\mathscr{P}_+ + \left(-\frac{\hbar}{2}\right)\mathscr{P}_-. \tag{5.2}$$

If we were to perform this experiment, we'd calculate the average using equation (5.1), but we can predict what the result will be from equation (5.2) as long as we know the state, since

$$\mathscr{P}_\pm = |\langle\pm|\psi\rangle|^2$$

as usual.

In fact, we can use this to write the calculation in a different way. We'll exploit the fact that the complex conjugate flips the bra and ket, so that

$$|\langle\pm|\psi\rangle|^2 = \langle\psi|\pm\rangle\langle\pm|\psi\rangle.$$

Equation (5.2) then becomes

$$\langle S_z\rangle = \left(+\frac{\hbar}{2}\right)\langle\psi|+\rangle\langle+|\psi\rangle + \left(-\frac{\hbar}{2}\right)\langle\psi|-\rangle\langle-|\psi\rangle.$$

Now here's a tricky part: since the factors of $\pm\hbar/2$ are just numbers, we can write them anywhere we want, so I'll move them into the middle of the first inner product in each term, getting

$$\langle S_z\rangle = \langle\psi|\left(+\frac{\hbar}{2}\right)|+\rangle\langle+|\psi\rangle + \langle\psi|\left(-\frac{\hbar}{2}\right)|-\rangle\langle-|\psi\rangle.$$

Next I'll replace those factors of $\hbar/2$ with the operator $\hat{S}_z$, since that operator acting on the ket to the right will produce the exact number we need:

$$\langle S_z\rangle = \langle\psi|\hat{S}_z|+\rangle\langle+|\psi\rangle + \langle\psi|\hat{S}_z|-\rangle\langle-|\psi\rangle.$$

Finally, notice that there are some common factors I can pull out (as long as I don't change the order of the operators and kets), so we can write

$$\langle S_z\rangle = \langle\psi|\hat{S}_z\Big(|+\rangle\langle+| + |-\rangle\langle-|\Big)|\psi\rangle.$$

But the thing inside the bracket is just the identity operator! I won't bother to write it since it doesn't change the ket it's acting on to the right, in which case we (finally) have

$$\langle S_z\rangle = \langle\psi|\hat{S}_z|\psi\rangle. \tag{5.3}$$

Evidently there are two different ways we can calculate the average of a series of measurements: using the possibilities and probabilities as in equation (5.2) or instead as in equation (5.3). In general, for any discrete system and for any observable A, we can write

$$\boxed{\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle = \sum_n a_n \mathscr{P}_{a_n}.} \tag{5.4}$$

In quantum mechanics, this average is called the *expectation value*.

Example 5.1 – Expectation Value.
Consider the state

$$|\psi\rangle = \frac{3}{5}|+\rangle + \frac{4}{5}|-\rangle.$$

Calculate $\langle S_z\rangle$ using both methods described above.

Using the possibilities and probabilities, we have

$$\langle S_z\rangle = \sum_n a_n \mathscr{P}_{a_n} = \left(+\frac{\hbar}{2}\right)\mathscr{P}_+ + \left(-\frac{\hbar}{2}\right)\mathscr{P}_-.$$

We can calculate the probabilities first,

$$\mathscr{P}_+ = |\langle+|\psi\rangle|^2 = 9/25 \quad \text{and} \quad \mathscr{P}_- = |\langle-|\psi\rangle|^2 = 16/25$$

so

$$\langle S_z\rangle = \left(+\frac{\hbar}{2}\right)\left(\frac{9}{25}\right) + \left(-\frac{\hbar}{2}\right)\left(\frac{16}{25}\right) = -\frac{7\hbar}{50} = -0.14\,\hbar.$$

The second method is more easily done in matrix notation:

$$\langle S_z\rangle = \langle\psi|\hat{S}_z|\psi\rangle \to \begin{pmatrix}3/5 & 4/5\end{pmatrix}\begin{pmatrix}\hbar/2 & 0\\ 0 & -\hbar/2\end{pmatrix}\begin{pmatrix}3/5\\ 4/5\end{pmatrix}.$$

If we multiply the operator by the ket first we get

$$\langle S_z\rangle = \langle\psi|\hat{S}_z|\psi\rangle \to \begin{pmatrix}3/5 & 4/5\end{pmatrix}\begin{pmatrix}3\hbar/10\\ -4\hbar/10\end{pmatrix} = -\frac{7\hbar}{50}.$$

5.2 Standard Deviation

If we run an experiment on a large ensemble of particles, we get a large number of results, all of which are random and in general all of which are potentially different (although for spin, they'll all be either $+\hbar/2$ or $-\hbar/2$). We've discussed above how we can find the average of all these results, but we can also calculate the standard deviation,

$$\sigma_A \equiv \sqrt{\langle(A-\langle A\rangle)^2\rangle}. \tag{5.5}$$

This is the square root of the average of the square of the deviations from the average, which sounds complicated but is a pretty common way to measure the spread of the results.

If we foil out the square, we can write this a little more conveniently:

$$\sigma_A = \sqrt{\langle A^2 - 2A\langle A\rangle + \langle A\rangle^2\rangle} = \sqrt{\langle A^2\rangle - 2\langle A\rangle\langle A\rangle + \langle A\rangle^2}$$

or

$$\sigma_A = \sqrt{\langle A^2\rangle - \langle A\rangle^2}. \tag{5.6}$$

Notice the difference between these two terms: the first is the average of the squares, and the second is the square of the averages. We already know how to calculate the average in quantum mechanics – we call it the expectation value – but what does $\langle A^2\rangle$ mean? Write it out:

$$\langle A^2\rangle = \langle\psi|\hat{A}^2|\psi\rangle = \langle\psi|\hat{A}\hat{A}|\psi\rangle.$$

We operate with the operator $\hat{A}$ *twice*.

In quantum mechanics, the standard deviation is often called the *uncertainty*. But be careful: this uncertainty is fundamental to the theory, and you can't design a better experiment to improve precision.

Example 5.2 – Standard Deviation.

Continuing with Example 5.1, find the standard deviation σ_{S_z}.

From the last example, we already have $\langle S_z\rangle = -7\hbar/50$, so we just need $\langle S_z^2\rangle$:

$$\langle S_z^2\rangle = \langle\psi|\hat{S}_z\hat{S}_z|\psi\rangle.$$

Writing this in matrix notation gives

$$\langle S_z^2 \rangle = \begin{pmatrix} 3/5 & 4/5 \end{pmatrix} \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

This is tedious but not difficult; I found that

$$\langle S_z^2 \rangle = \frac{\hbar^2}{4},$$

so the standard deviation is

$$\begin{aligned} \sigma_{S_z} &= \sqrt{\langle S_z^2 \rangle - \langle S_z \rangle^2} \\ &= \sqrt{\hbar^2/4 - 49\hbar^2/2500} \\ &= \frac{24}{50}\hbar. \end{aligned}$$

5.3 Commuting Operators

Although the order in which you multiply two numbers doesn't matter – $6 \times 7 = 7 \times 6$ – you probably know that for matrices the order *does* matter. We say that matrices don't *commute*. As you might expect, the same is true for operators as well. For example, try $\hat{S}_x$ and $\hat{S}_z$:

$$\hat{S}_x \hat{S}_z \to \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix} \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

and

$$\hat{S}_z \hat{S}_x \to \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

It's pretty clear $\hat{S}_z \hat{S}_x \neq \hat{S}_x \hat{S}_z$.

Because most operators in quantum mechanics don't commute with each other, and it means something special if two operators *do* in fact commute, we define a new calculation called the *commutator*. The commutator for two operators $\hat{A}$ and $\hat{B}$ is

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{5.7}$$

You can think of it as a measure of how much two operators *don't* commute, and of course if $[\hat{A}, \hat{B}] = 0$, then $\hat{A}$ and $\hat{B}$ actually do commute with each other.

Now, I mentioned that it means something special if two operators commute, but the argument is a little subtle so read carefully. Suppose $\hat{A}$ and $\hat{B}$ commute, so that $\hat{A}\hat{B} = \hat{B}\hat{A}$ and we can apply them to a state in any order and get the same result,

$$\hat{A}\hat{B}|\psi\rangle = \hat{B}\hat{A}|\psi\rangle.$$

But now suppose that the state we're acting on is an eigenstate of operator $\hat{A}$, which I'll write as $|a\rangle$, with eigenvalue a:

$$\hat{A}|a\rangle = a|a\rangle.$$

Let's now act on both sides of this equation with operator $\hat{B}$,

$$\hat{B}\Big(\hat{A}|a\rangle\Big) = \hat{B}\big(a|a\rangle\big).$$

But the a on the right hand side is just a number, and although we have to be careful moving operators around (they don't always commute), numbers can be put anywhere. So I'll move the a to the other side of $\hat{B}$. While I'm doing that, I'll also switch the order of the operators on the left hand side, so that we have

$$\hat{A}\hat{B}|a\rangle = a\hat{B}|a\rangle.$$

I'll put in some suggestive brackets to guide your eye:

$$\hat{A}\Big(\hat{B}|a\rangle\Big) = a\Big(\hat{B}|a\rangle\Big).$$

As you can see, the same thing is in each bracketed term; what is it? Well, remember that when an operator ($\hat{B}$ in this case) acts on a state, a new state is produced, so the brackets enclose a state. But since acting on that state with the operator $\hat{A}$ gives you back that same state, it must also be an eigenstate of $\hat{A}$; that is, both $|a\rangle$ and $\hat{B}|a\rangle$ are eigenstates of the operator $\hat{A}$, with the same eigenvalue a. This means that the state $\hat{B}|a\rangle$ must be a scalar multiple of the state $|a\rangle$:

$$\hat{B}|a\rangle = b|a\rangle.$$

Our conclusion is that if $\hat{A}$ and $\hat{B}$ commute, they have common (sometimes called simultaneous) eigenstates. They'll have different eigenvalues, though.

If this is the case – that $\hat{A}$ and $\hat{B}$ commute – we say that they are *compatible observables*. Why? Because making a measurement of one observable won't affect a measurement of the other. Let's run through this: suppose we start in some general state $|\psi\rangle$, and we first measure the

observable A. We get one of the eigenvalues of the operator $\hat{A}$ – call it a_n – and the state collapses to $|a_n\rangle$. Next, make a measurement of B. Here's the thing: the current state is also an eigenstate of $\hat{B}$, so we necessarily measure b_n – the eigenvalue of $\hat{B}$ that corresponds to the state $|a_n\rangle$. But this time the state doesn't collapse; it stays in $|a_n\rangle$ since it's the eigenstate. That means a subsequent measurement of A will also return the value a_n. The observables A and B are compatible because measuring one doesn't cause us to lose information about the other.

Example 5.3 – Commutators.
What is the commutator of $\hat{S}_z$ and $\hat{S}_x$?

We already calculated $\hat{S}_x\hat{S}_z$ and $\hat{S}_z\hat{S}_x$ earlier, so we can subtract them to find the commutator:

$$[\hat{S}_z, \hat{S}_x] = \hat{S}_z\hat{S}_x - \hat{S}_x\hat{S}_z \to \frac{\hbar^2}{4}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \frac{\hbar^2}{4}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
$$= \frac{\hbar^2}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

I don't know if that matrix in the answer looks familiar to you yet, but it's pretty close to the operator $\hat{S}_y$. In fact, we can write

$$[\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y. \tag{5.8}$$

Similarly,

$$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z. \tag{5.9}$$

and

$$[\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x. \tag{5.10}$$

5.4 The Uncertainty Principle

You might already know Heisenberg's famous uncertainty principle, usually written as

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

It means, in brief, that the more certain you are of position x, they less certain you can be of your momentum p, and vice versa. In fact, this uncertainty principle is just one example of a more general uncertainty

principle, which I'll give you here without proof:

$$\boxed{\sigma_A \sigma_B \geq \frac{1}{2}\left|\langle [\hat{A}, \hat{B}] \rangle\right|.} \tag{5.11}$$

Notice that I'm using the standard deviation explicitly – that's what the uncertainty really is, after all – even though it's a little nonstandard in the field.

What does this mean, exactly? Well, we already know that two operators that don't commute have incompatible observables – measuring one will affect your knowledge of the second. The uncertainty principle quantifies this for us, and gives us the relationship between the spread of values we would get upon measuring A or B.

Example 5.4 – Uncertainty Principle.

Suppose our two operators are $\hat{S}_x$ and $\hat{S}_y$. We already know they don't commute, and in fact

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z.$$

Plugging this into equation (5.11) gives

$$\sigma_{S_x} \sigma_{S_y} \geq \frac{\hbar}{2}\left|\langle S_z \rangle\right|.$$

If we know what state we're in, we can also figure out $\langle S_z \rangle$; if, for example, we're in state $|+\rangle$, then every measurement returns $\hbar/2$ so $\langle S_z \rangle = \hbar/2$, and the uncertainty principle now reads

$$\sigma_{S_x} \sigma_{S_y} \geq \frac{\hbar^2}{4}.$$

This is a mathematical statement of something we've already seen: if we know exactly what S_z is, we can't know either S_x or S_y perfectly – neither can have a definite value, since then their standard deviation would be zero.

Problems

Problem 5.1 – Expectation Value Practice.
In your lab you create a beam of particles, all in the state

$$|\psi\rangle = \frac{2}{\sqrt{13}}|+\rangle + \frac{3i}{\sqrt{13}}|-\rangle.$$

(a) Calculate the expectation value $\langle S_z \rangle$.

(b) Calculate the expectation value $\langle S_x \rangle$.

(c) Calculate the expectation value $\langle S_y \rangle$.

(d) Calculate the expectation value $\langle S^2 \rangle$.

(e) If you were to use 100 particles to measure these quantities experimentally, would you expect your results to exactly match your theoretical predictions above? Why or why not?

Problem 5.2 – More Expectation Value Practice.
In your lab you create a beam of particles, all in the state

$$|\psi\rangle = |a_1\rangle - 2|a_2\rangle + 5|a_3\rangle$$

(this is the three-state system from Problems 2.2 and 3.7, where a_1, a_2, and a_3 correspond to the three possible results of measuring observable A). What is the expectation value $\langle A \rangle$?

Problem 5.3 – Standard Deviations. Continuing from Problem 5.1, calculate the following:

(a) The standard deviation σ_{S_z}.

(b) The standard deviation σ_{S_x}.

(c) The standard deviation σ_{S_y}.

(d) The standard deviation σ_{S^2}.

Problem 5.4 – Uncertainty Principle. Continuing from Problems 5.1 and 5.3, show that the uncertainty principle for $\hat{S}_x$ and $\hat{S}_y$ is satisfied (that is, show that the left hand side of the uncertainty principle is greater than or equal to the right hand side).

Chapter 6

The Schrödinger Equation

6.1 The Hamiltonian and the Energy Basis

Let's get right into it: the time evolution of a quantum system is determined by the Schrödinger equation,

$$\boxed{i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle.} \tag{6.1}$$

There's a lot to unpack here, though. First of all, notice that this is a differential equation for the state $|\psi(t)\rangle$, and I've now explicitly put in the time dependence of the state; from this point on, we should probably assume all states will be time dependent.

Second, we've introduced a new operator: the Hamiltonian $\hat{H}(t)$. This is the quantum operator corresponding to the classical Hamiltonian – the total energy of the system, kinetic energy plus potential energy:[1]

$$H = T + U. \tag{6.2}$$

The Hamiltonian operator is by far the most important operator in quantum mechanics, and not just because of its appearance in the Schrödinger equation. So let's dig a little deeper into it.

A big part of our job going forward in the book will be to determine the eigenvalues and eigenvectors of the Hamiltonian operator; we'll write the eigenvalue equation as

$$\boxed{\hat{H}|E_n\rangle = E_n|E_n\rangle.} \tag{6.3}$$

[1] You might remember from your classical mechanics course that the Hamiltonian isn't always equal to the total energy of the system, but we won't have to worry about that in this book.

The notation here is a little different than for other operators; I've written the eigenvalues as E_n (where n is an index – there will usually be more than one eigenvalue) and the eigenkets as $|E_n\rangle$. That's because the Hamiltonian corresponds to the observable energy: when we measure H, we're really measuring the total energy of the system. The eigenvalues E_n are often called the *allowed energies*. Much of quantum mechanics at this stage is a pursuit of those allowed energies for different systems.

Since it represents an observable, the Hamiltonian must be a Hermitian operator. As I mentioned back in Section 4.8, the eigenvalues of a Hermitian operator are real – which is nice since in this case they're they're energies, and it would be strange if we had imaginary energy. Furthermore, the eigenstates are orthonormal,

$$\langle E_m|E_n\rangle = \delta_{mn}, \tag{6.4}$$

and complete, meaning *any* state can be written as a superposition of them,

$$|\psi(t)\rangle = c_1|E_1\rangle + c_2|E_2\rangle + c_3|E_3\rangle + \cdots = \sum_n c_n(t)\,|E_n(t)\rangle. \tag{6.5}$$

If the Hamiltonian is *time independent*, then we'd expect the eigenstates $|E_n\rangle$ to also be time independent. But that doesn't mean the state $|\psi(t)\rangle$ doesn't evolve in time – in general it will. In this case all the time dependence is carried in the coefficients $c_n(t)$, and it turns out we can solve the Schrödinger equation for any system – as long as the Hamiltonian of that system doesn't depend on time.

The trick is to write the state in the energy basis, as in equation (6.5). If we plug that into the Schrödinger equation, we get

$$i\hbar\frac{d}{dt}\left(\sum_n c_n(t)|E_n\rangle\right) = \hat{H}\left(\sum_n c_n(t)|E_n\rangle\right).$$

Now we can move the derivative into the sum, and since the eigenstate $|E_n\rangle$ doesn't depend on time here, it will only act on the coefficients $c_n(t)$. On the right hand side, we can move the Hamiltonian operator right beside the state, since the coefficients are just numbers, giving

$$i\hbar\sum_n \frac{dc_n}{dt}|E_n\rangle = \sum_n c_n(t)\hat{H}|E_n\rangle.$$

Of course, we know that $\hat{H}|E_n\rangle = E_n|E_n\rangle$ – it's the eigenstate of the operator, after all, and the reason why we are using the energy basis here in

the first place, so we have

$$i\hbar \sum_n \frac{dc_n}{dt}|E_n\rangle = \sum_n c_n(t)E_n|E_n\rangle.$$

We need to solve this equation for the unknown coefficients $c_n(t)$, but they're buried in the sums. To get rid of the sums and isolate the coefficeints, we take the inner product with the eigenstate $|E_m\rangle$ – notice the index is m rather than n:

$$\langle E_m|\left(i\hbar \sum_n \frac{dc_n}{dt}|E_n\rangle\right) = \langle E_m|\left(\sum_n c_n(t)E_n|E_n\rangle\right).$$

Now let the bra act on the ket on each side to get

$$i\hbar \sum_n \frac{dc_n}{dt}\langle E_m|E_n\rangle = \sum_n c_n(t)E_n\langle E_m|E_n\rangle.$$

But since the eigenstates are orthonormal, we have

$$\langle E_m|E_n\rangle = \delta_{mn}.$$

This is a little tricky to see at first, but that Kronecker delta δ_{mn} will collapse the sum – every term in the sum will end up being zero except for the mth one. So the equation reduces to

$$i\hbar \frac{dc_m}{dt} = c_m E_m. \tag{6.6}$$

Finally, this is differential equation we can readily solve; separating and integrating gives the solution as

$$c_m(t) = c_{m,0}e^{-iE_m t/\hbar} \tag{6.7}$$

(try it yourself!), where $c_{m,0} = c_m(0)$.

We have now the solution to the Schrödinger equation in cases where the Hamiltonian is time independent. There's just one last thing: it's customary to drop the naught in the initial value of the coefficient $c_{m,0}$ and just write it as c_m (or, as I'll write it in a moment, c_n – the index letter doesn't matter). This might seem a little confusing, since c_m could also stand for the value at time t, $c_m(t)$, but it'll turn out we'll never write it that way – c_m will henceforth always mean the initial value.

Anyway, here we go: the solution to the Schrödinger equation, written in the energy basis, is

$$\boxed{|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar}|E_n\rangle.} \tag{6.8}$$

And, just to be clear, we're not *really* done – to finish this off, we need to know the energy eigenstates $|E_n\rangle$ (which will depend on the Hamiltonian $\hat{H}$) and all of the coefficients $c_n = c_n(0)$ (which will depend on the initial conditions).

6.2 Stationary States

Although we'll be spending the next several chapters solving the energy eigenvalue equation – really, we'll be doing it *a lot* – let's pretend for a moment that we already know what the eigenvalues E_n and eigenstates $|E_n\rangle$ are, and suppose that our system just happens to initially be in the state $n = 1$ state:

$$|\psi(0)\rangle = |E_1\rangle.$$

This is the energy eigenstate, so we know that a measurement of the energy on the system at time $t = 0$ will yield $E = E_1$ (whatever that value happens to be). Question: what will a measurement of the energy get at a later time t?

This is what the Schrödinger equation does: given an initial condition, it tells us how the state evolves in time, and the answer is apparently given by equation (6.8). We still have to figure out all the coefficients c_n, but we can do that by using our initial condition. Plugging in $t = 0$ into equation (6.8) gives

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle.$$

But we know $|\psi(0)\rangle = |E_1\rangle$, so we must have $c_1 = 1$ and every other $c_n = 0$. Then we just go back to the full solution and plug in our known coefficients c_n to get the first part of our answer: the state at a later time t is given by

$$|\psi(t)\rangle = e^{-iE_1 t/\hbar}|E_1\rangle.$$

Let's take a look at this state. Notice that the only difference from the initial state is the presence of the exponential factor in front – but that's an overall phase factor, which we've said doesn't physically change the state. So has anything really changed in time? Not really; if we measure the energy, the probability we'll get E_1 is

$$\mathscr{P}_{E_1} = |\langle E_1|\psi(t)\rangle|^2 = |e^{-iE_1 t/\hbar}\langle E_1|E_1\rangle|^2 = |e^{-iE_1 t/\hbar}|^2 = 1,$$

or 100% – the same as at $t = 0$.

In fact, *any* measurement of *any* observable – which I'll write as A – is time-independent:

$$\mathscr{P}_{a_n} = |\langle a_n|\psi(t)\rangle|^2 = |e^{-iE_1 t/\hbar}\langle a_n|E_1\rangle|^2 = |\langle a_n|E_1\rangle|^2.$$

This probability obviously depends on the observable A that we're measuring, but not on time at all. States for which there is no measurable time evolution – that is, every measurement we can make has no time dependence – are called *stationary states*. The energy eigenstates of a system are always stationary states.

6.3 Superposition States

What if our initial state was

$$|\psi(0)\rangle = c_1|E_1\rangle + c_2|E_2\rangle$$

instead? This is called a *superposition state* for obvious reasons. In this case equation (6.8), after applying our initial conditions, gives us the state at a later time as

$$|\psi(t)\rangle = c_1 e^{-iE_1 t/\hbar}|E_1\rangle + c_2 e^{-iE_2 t/\hbar}|E_2\rangle.$$

You can try using the initial conditions to show this if you like, but notice the pattern that's forming: if we write our initial state in the energy basis, then all we have to do to evolve that state forward in time is to "tack on" the exponential time factors $e^{-iE_n t/\hbar}$ to each appropriate term.

If we measure the energy of our system at some later time t, notice that the probabilities stay constant in time:

$$\mathscr{P}_{E_1} = |\langle E_1|\psi(t)\rangle|^2 = |c_1|^2, \quad \text{and} \quad \mathscr{P}_{E_2} = |c_2|^2.$$

This is *always* the case if the Hamiltonian is time independent – energy probabilities don't change.

What about some other observable A? Then it depends. If the operators $\hat{A}$ and $\hat{H}$ commute, then they share eigenstates – so the states $|E_n\rangle$ are also eigenstates of $\hat{A}$ and the probabilities will be constant as well.

If $\hat{A}$ and $\hat{H}$ *don't* commute, they won't share eigenstates. However, since the energy eigenstates are complete, we could always write the eigenstates of $\hat{A}$ as superpositions of the energy eigenstates,

$$|a_m\rangle = \sum_n \alpha_n |E_n\rangle. \tag{6.9}$$

Then we can calculate measurement probabilities $\mathscr{P}_{a_n}$, but they would end up varying with time.

This last part is a little confusing, so let's go through the process with a simple example.

Example 6.1 – Time Evolution.

Consider some generic two-state system – we'll do some more specific and physical examples in the next chapter – with a Hamiltonian given by

$$\hat{H} \to \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}.$$

Notice that, as a diagonal matrix, $\hat{H}$ is written here in its own basis (the energy basis), and that means the diagonal terms E_1 and E_2 are the possible energies we'd measure.

Suppose we have a second observable $\hat{A}$ which we'll write in the energy basis as

$$\hat{A} \to \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}.$$

We're going to explore this system in detail, and it starts with finding the eigenvalues and eigenstates of these two operators. Luckily we already have the answer for the Hamiltonian: the eigenvalues are the allowed energies E_1 and E_2, and the associated eigenstates are just the kets $|E_1\rangle$ and $|E_2\rangle$, which we'll write in matrix notation as

$$|E_1\rangle \to \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |E_2\rangle \to \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The operator $\hat{A}$ requires more work. We have to solve the characteristic equation, $\det|\mathbf{A}-\lambda\mathbf{I}| = 0$, where λ represents the possible eigenvalues. Let's work it out:

$$\begin{vmatrix} -\lambda & a \\ a & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - a^2 = 0,$$

so $\lambda = \pm a$. To find the eigenstates, we need to write down the eigenvalue equation,

$$\hat{A}|a\rangle = \lambda|a\rangle,$$

and plug in our eigenvalues, one at a time. For $\lambda = +a$, we can write

$$\hat{A}|+a\rangle = +a|+a\rangle$$

and solve for the ket $|+a\rangle$. This is easy to do in matrix notation, where the eigenvalue equation becomes

$$\begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = a\begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Multiplying out the left hand side and equating the top rows of each side (or the bottom rows) gives $\beta = \alpha$, so we know our eigenstate is

$$\begin{pmatrix}\alpha\\ \alpha\end{pmatrix}.$$

But how do we get α? Remember, every state must be normalized, so we must also have

$$\langle +a|+a\rangle = 1 \rightarrow \begin{pmatrix}\alpha & \alpha\end{pmatrix}\begin{pmatrix}\alpha\\ \alpha\end{pmatrix} = 1,$$

or $\alpha = 1/\sqrt{2}$. Our eigenstate $|+a\rangle$ is

$$|+a\rangle = \frac{1}{\sqrt{2}}|E_1\rangle + \frac{1}{\sqrt{2}}|E_2\rangle \rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ 1\end{pmatrix}.$$

Similarly – try it on your own – the eigenstate $|-a\rangle$, corresponding to eigenvalue $-a$, is

$$|-a\rangle = \frac{1}{\sqrt{2}}|E_1\rangle - \frac{1}{\sqrt{2}}|E_2\rangle \rightarrow \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -1\end{pmatrix}.$$

Well, that's been a lot of work already, and we haven't even really answered any questions. Let's answer some now. We'll set up our system so that at time $t = 0$ it's in the state

$$|\psi(0)\rangle = |+a\rangle.$$

(a) A measurement of the observable A is made at $t = 0$. What are the possible results and probabilities of each?

This one is easy since we start out in an eigenstate of $\hat{A}$ – we'll measure $A = +a$ with a probability of 100%, and never measure $A = -a$.

(b) A measurement of the energy E is made at $t = 0$. What are the possible results and probabilities of each?

The possibilities are the energy eigenvalues – the allowed energies – which are E_1 and E_2. The probability of measuring $E = E_1$ can be calculated in the usual way,

$$\mathscr{P}_{E_1} = |\langle E_1|\psi(0)\rangle|^2 = |\langle E_1|+a\rangle|^2 \rightarrow \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{pmatrix}\right|^2 = \frac{1}{2}.$$

Of course, that means we must have $\mathscr{P}_{E_2} = 1/2$ as well. So we have a 50/50 chance of measuring each allowed energy.

(c) What is the state at a later time?

To write out the state at a later time, we have to make two steps: first, write the initial state in the energy basis. We already have this from above,

$$|+a\rangle = \frac{1}{\sqrt{2}}|E_1\rangle + \frac{1}{\sqrt{2}}|E_2\rangle.$$

Second, tack on the appropriate exponential time factors, giving

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-iE_1t/\hbar}|E_1\rangle + \frac{1}{\sqrt{2}}e^{-iE_2t/\hbar}|E_2\rangle.$$

That's it!

(d) A measurement of the energy E is made at a time t. What are the possible results and probabilities of each?

The possible results are still the allowed energies E_1 and E_2. What about the probabilities? Well,

$$\mathscr{P}_{E_1} = |\langle E_1|\psi(t)\rangle|^2 = \left|\frac{1}{\sqrt{2}}e^{-iE_1t/\hbar}\right|^2 = \frac{1}{2}.$$

Then $\mathscr{P}_{E_2} = 1/2$ as well – but these are the same values we found at $t = 0$. As promised, the energy probabilities are constant in time.

(e) A measurement of the observable A is made at a time t. What is the probability of measuring $A = +a$?

Let's make the calculation in matrix notation, but be careful to make sure everything is in the same basis – the energy basis in this case. Also, it gets a little messy:

$$\begin{aligned}\mathscr{P}_{+a} &= |\langle +a|\psi(t)\rangle|^2 \\ &= \left|\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}e^{-iE_1t/\hbar} \\ e^{-iE_2t/\hbar}\end{pmatrix}\right|^2 \\ &= \frac{1}{4}\left|e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\right|^2\end{aligned}$$

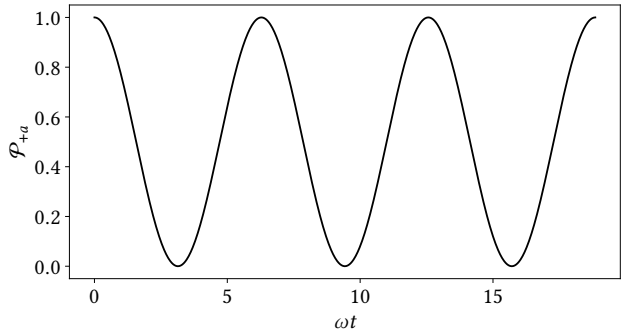


Figure 6.1: The probability of measuring $A = +a$. It varies sinusoidally with time.

Remember that those absolute value symbols actually mean we have to multiply by the complex conjugate, so

$$\begin{aligned}\mathcal{P}_{+a} &= \frac{1}{4}\Big(e^{iE_1t/\hbar} + e^{iE_2t/\hbar}\Big)\Big(e^{-iE_1t/\hbar} + e^{-iE_2t/\hbar}\Big) \\ &= \frac{1}{4}\Big(2 + e^{i(E_1-E_2)t/\hbar} + e^{i(E_2-E_1)t/\hbar}\Big) \\ &= \frac{1}{4}\Big(2 + e^{i\omega t} + e^{-i\omega t}\Big) \\ &= \frac{1}{2}\Big(1 + \cos(\omega t)\Big),\end{aligned}$$

where I've defined $\omega \equiv (E_2 - E_1)/\hbar$. Notice that ω has the units of angular frequency – loosely, it tells us how fast the state is changing in time. Anyway, now we have an interesting situation – something finally changes in time! The probability of measuring $A = +a$ varies with time; at $t = 0$ it's 100%, but at time $t = \pi/\omega$ it's actually zero. At time $t = 2\pi/\omega$ it's back up to 100%, and in fact it oscillates sinusoidally in time as shown in Figure 6.1.

Problems

Problem 6.1 – Solving the Schrödinger Equation.

In Section 6.1 we solved the Schrödinger equation for the case where the Hamiltonian doesn't depend on time, and we got down to the differ-

ential equation (equation 6.6)

$$i\hbar \frac{dc_m}{dt} = c_m E_m.$$

Solve this equation and show that that the answer is as given in equation (6.7).

Problem 6.2 – Checking the Solution.

Check our solution to the Schrödinger equation (equation 6.8) by explicitly plugging it into the Schrödinger equation itself and showing that the right hand side equals the left hand side.

Problem 6.3 – Practice 1.

Consider the Hamiltonian given by

$$\hat{H} \to E_0 \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

where E_0 is some real positive constant. This matrix is written in the basis given by the kets $|q_1\rangle$ and $|q_2\rangle$, which are eigenstates of some operator $\hat{Q}$, and have corresponding eigenvalues q_1 and q_2. Initially the system starts out in a state given by

$$|\psi(0)\rangle = \frac{3}{5}|q_1\rangle + \frac{4}{5}|q_2\rangle.$$

(a) What are the allowed energies of the system?

(b) Find the energy eigenstates of the Hamiltonian.

(c) If you were to make a measurement of the energy of the system at $t = 0$, what are the probabilities of measuring each?

(d) If you were to measure Q at time $t = 0$, what are the possible results and probabilities of each?

(e) What is the state $|\psi(t)\rangle$ at a later time?

(f) If you were to measure Q at a later time t, what are the possible results and probabilities of each?

(g) What is the expectation value $\langle A(t)\rangle$ as a function of time?

Problem 6.4 – Practice 2.
Consider the Hamiltonian given by

$$\hat{H} \to \begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix},$$

where E_0, E_1, and A are real positive constants. This matrix is written in a basis given by the kets $|1\rangle$, $|2\rangle$, and $|3\rangle$.

(a) Find the energy eigenvalues and eigenstates.

(b) Write the Hamiltonian in the energy basis (as a matrix).

(c) Suppose the initial state of the system is $|\psi(0)\rangle = |1\rangle$. What is the probability that the system is in state $|1\rangle$ at time t?

(d) Suppose the initial state of the system is $|\psi(0)\rangle = |2\rangle$. What is the probability that the system is in state $|2\rangle$ at time t?

Chapter 7

Examples of Discrete Systems

With the exception of Chapter 1, we haven't dealt deeply with any realistic physical systems, but now that all the rules are on the table we can turn to applying them to a few interesting systems. We'll stick with *two-state systems* for mathematical simplicity, but there are interesting three-state systems (or higher) and all the same rules apply there, too.

7.1 An Electron in a Magnetic Field

Back in Chapter 1 we discussed one of the implications of electrons (or any other spin-1/2 particles) having an intrinsic angular momentum $\mathbf{S}$ – it's spin – namely, that a particle is deflected by an inhomogenous magnetic field. However, there's another effect as well: the magnetic field will actually change the spin vector of the electron. To see how this happens, we need to figure out the Hamiltonian of the system and see how it evolves according to the Schrödinger equation.

Consider an electron at rest in a constant (and homogeneous this time) magnetic field aligned along the z-axis and given by

$$\mathbf{B} = B_0 \hat{z}, \tag{7.1}$$

as shown in Figure 7.1. The spin of the electron gives it a magnetic dipole moment $\boldsymbol{\mu}$, given in equation (1.3) for an electron as

$$\boldsymbol{\mu} = -\frac{e}{m}\mathbf{S}.$$

Classically, the magnetic field would apply a torque

$$\mathbf{N} = \boldsymbol{\mu} \times \mathbf{B}$$

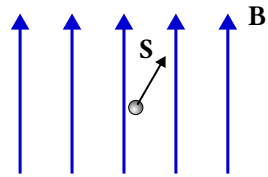


Figure 7.1: An electron in a magnetic field.

to the dipole, causing it to try to align with the magnetic field.[1] But to understand the quantum system, we need to work with energies rather than forces, and the potential energy of a dipole in a magnetic field is given by

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}. \tag{7.2}$$

Since our electron is at rest and has no kinetic energy, this is also the full Hamiltonian of our system,

$$H = -\boldsymbol{\mu} \cdot \mathbf{B} = \frac{e}{m}\mathbf{S} \cdot \mathbf{B} = \omega_0 S_z, \tag{7.3}$$

where $\omega_0 \equiv eB_0/m$ is a constant – we'll see later what it means.

To "turn on quantum mechanics" we just turn our observable into an operator, like so:

$$\hat{H} = \omega_0 \hat{S}_z \tag{7.4}$$

(that's right, I just put hats on the observables). Remember, this Hamiltonian does double duty: it tells us the allowed energies and associated energy eigenstates, and it also tells us, within the Schrödinger equation, how a state in this system evolves in time.

Let's do the first thing first, since it's pretty easy: we need to find the eigenvalues and eigenstates of the Hamiltonian operator. That's actually trivial in this case, since the Hamiltonian is just proportional to the spin operator $\hat{S}_z$; writing the Hamiltonian in matrix notation (in the S_z basis as usual),

$$\hat{H} \to \frac{\hbar\omega_0}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we can see that it's diagonal and therefore the eigenvalues must be

$$E_+ = \frac{\hbar\omega_0}{2} \quad \text{and} \quad E_- = -\frac{\hbar\omega_0}{2}. \tag{7.5}$$

[1] See, for example, Griffiths *Introduction to Electrodynamics*, section 6.1.2.

The eigenstates are even easier; they're the same as the basis states, so

$$|E_+\rangle = |+\rangle \quad \text{and} \quad |E_-\rangle = |-\rangle. \tag{7.6}$$

Now that we have the Hamiltonian all figured out, we can answer some basic questions about what happens to an electron in a magnetic field. Let's suppose the electron is initially in some general spin state,

$$|\psi(0)\rangle = a|+\rangle + b|-\rangle.$$

Actually, it's a little more convenient to write the coefficients as

$$a = \cos(\theta/2) \quad \text{and} \quad b = \sin(\theta/2);$$

we'll see what the angle θ means physically in bit, but notice for now that writing the state as

$$|\psi(0)\rangle = \cos(\theta/2)|+\rangle + \sin(\theta/2)|-\rangle \tag{7.7}$$

isn't any less general than using a and b; there's always some angle that will reproduce whatever values we wanted. And, as a bonus, the state is automatically normalized, since

$$|a|^2 + |b|^2 = \cos^2(\theta/2) + \sin^2(\theta/2) = 1.$$

Here's our first question: given that initial state, what is the state at a later time? We can answer this with our solution to the Schrödinger equation discussed last chapter, and which will always take the following two steps:

1. put the initial state into the energy basis, and
2. tack on the exponential time factors $e^{-iE_n t/\hbar}$ to each energy basis state.

In this case, the first step is already done for us – the energy basis is the spin S_z basis. The second step gives us

$$|\psi(t)\rangle = \cos(\theta/2)e^{-iE_+ t/\hbar}|+\rangle + \sin(\theta/2)e^{-iE_- t/\hbar}|-\rangle,$$

or, filling in our known energies $E_\pm$,

$$|\psi(t)\rangle = \cos(\theta/2)e^{-i\omega_0 t/2}|+\rangle + \sin(\theta/2)e^{i\omega_0 t/2}|-\rangle.$$

One more little trick – a pretty common one in quantum mechanics. Let's pull out a factor of $e^{-i\omega_0 t/2}$ from each term to get

$$|\psi(t)\rangle = e^{-i\omega_0 t/2}\Big[\cos(\theta/2)|+\rangle + \sin(\theta/2)e^{i\omega_0 t}|-\rangle\Big].$$

Why do this? Well, that first exponential is an overall phase factor, which we're allowed to ignore, so our final answer is the slightly nicer looking state

$$|\psi(t)\rangle = \cos(\theta/2)|+\rangle + \sin(\theta/2)e^{i\omega_0 t}|-\rangle. \tag{7.8}$$

This is the state at any later time t.

Suppose we want to measure the energy of our state at time t – what will we get? Well, we know the possible results are the eigenvalues of the Hamiltonian, and probabilities are calculated in the usual way, so

- we could measure $E_+ = \hbar\omega_0/2$ with a probability $\mathscr{P}_{E_+} = |\langle+|\psi(t)\rangle|^2 = \cos^2(\theta/2)$, or
- we could measure $E_- = -\hbar\omega_0/2$ with a probability $\mathscr{P}_{E_-} = |\langle-|\psi(t)\rangle|^2 = |\sin(\theta/2)e^{i\omega_0 t}|^2 = \sin^2(\theta/2)$.

As expected, these probabilities are constant – they won't change with time as the state evolves.

To see some more interesting time behaviour, we need to measure something other than energy. Let's try the spin component S_x. This is spin, so the two possible results are as usual $\pm\hbar/2$. Let's calculate the probability of measuring spin up along x:

$$\begin{aligned}\mathscr{P}_{+x} &= |{}_x\langle+|\psi(t)\rangle|^2 \\ &= \left|\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\begin{pmatrix}\cos(\theta/2)\\ \sin(\theta/2)e^{i\omega_0 t}\end{pmatrix}\right|^2.\end{aligned}$$

Multiplying the matrices gives

$$\mathscr{P}_{+x} = \frac{1}{2}\left|\cos(\theta/2) + \sin(\theta/2)e^{i\omega_0 t}\right|^2,$$

but be careful with those absolute value signs. In quantum mechanics they always mean multiply by the complex conjugate, so

$$\mathscr{P}_{+x} = \frac{1}{2}\Big(\cos(\theta/2) + \sin(\theta/2)e^{-i\omega_0 t}\Big)\Big(\cos(\theta/2) + \sin(\theta/2)e^{i\omega_0 t}\Big),$$

which simplifies to

$$\mathscr{P}_{+x} = \frac{1}{2}\Big[1 + \cos(\theta/2)\sin(\theta/2)\Big(e^{i\omega_0 t} + e^{-i\omega_0 t}\Big)\Big].$$

We can use the identity $\sin\theta = 2\cos(\theta/2)\sin(\theta/2)$ as well as Euler's formula for cosine,

$$e^{i\omega_0 t} + e^{-i\omega_0 t} = 2\cos\omega_0 t,$$

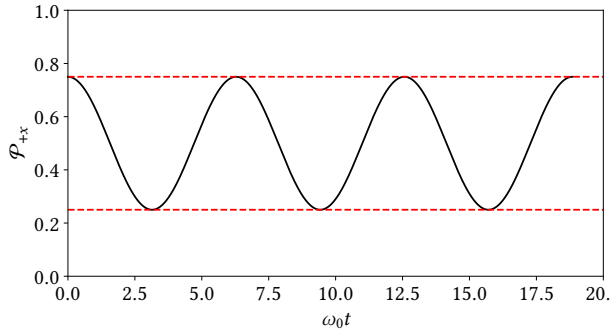


Figure 7.2: The probability $\mathcal{P}_{+x}$ of measuring the electron to have $S_x = \hbar/2$ as the state evolves in time. The parameter $\theta = \pi/6$ in this case.

to simplify this further, getting finally

$$\mathcal{P}_{+x} = \frac{1}{2}\Big(1 + \sin\theta\cos\omega_0 t\Big). \tag{7.9}$$

That was a lot of work, but the result is interesting: it's time dependent, and I've plotting the probability as a function of time in Figure 7.2. Note that the probability alternates between the values $(1 - \cos\theta)/2$ and $(1 + \cos\theta)/2$.

While we're looking at spin along the x direction, let's calculate the expectation value as well. It's pretty straightforward in matrix notation, although tedious:

$$\begin{aligned}\langle S_x(t)\rangle &= \langle\psi(t)|\hat{S}_x|\psi(t)\rangle \\ &\to \begin{pmatrix}\cos(\theta/2) & \sin(\theta/2)e^{-i\omega_0 t}\end{pmatrix}\begin{pmatrix}0 & \hbar/2 \\ \hbar/2 & 0\end{pmatrix}\begin{pmatrix}\cos(\theta/2) \\ \sin(\theta/2)e^{i\omega_0 t}\end{pmatrix},\end{aligned}$$

which evaluates to (with some work)

$$\langle S_x(t)\rangle = \frac{\hbar}{2}\sin\theta\cos\omega_0 t. \tag{7.10}$$

If we instead calculate the expectation value for S_y, we would get the similar

$$\langle S_y(t)\rangle = -\frac{\hbar}{2}\sin\theta\sin\omega_0 t, \tag{7.11}$$

and if we did S_z, we'd get

$$\langle S_z(t)\rangle = \frac{\hbar}{2}\cos\theta. \tag{7.12}$$

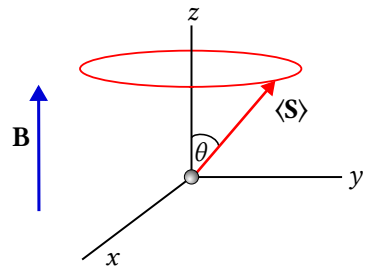


Figure 7.3: The expectation value of the spin of the electron in a magnetic field precesses around the z axis.

Not surprisingly, this last expectation value is constant since $\hat{S}_z$ and $\hat{H}$ commute.

This all actually makes a pretty clear physical picture of what's happening to the spin of the electron in the magnetic field. Consider the quantity

$$\langle \mathbf{S} \rangle = \langle S_x \rangle \hat{x} + \langle S_y \rangle \hat{y} + \langle S_z \rangle \hat{z}, \tag{7.13}$$

which in a way might represent the average direction the spin vector points. With the time behaviour in the x and y components of the vector, it looks like it precesses around the z axis, as shown in Figure 7.3. The precession frequency is

$$\omega_0 = \frac{eB_0}{m},$$

and the angle that $\langle \mathbf{S} \rangle$ makes with the z axis is the angle θ that we introduced way back at the start of the problem.

Interestingly, *classical* theory predicts the exact same thing for the magnetic dipole moment of the electron, a phenomenon called *Larmar precession*, and the classical frequency is the same as our ω_0, called the Larmar frequency. The only difference between classical theory and our result here is the presence of the expectation value rather than the actual physical observable. It turns out this is a general thing, and we'll encounter it many times; it's called Ehrenfest's theorem:

Ehrenfest's theorem: quantum expectation values obey classical laws.

7.2 Neutrino Oscillations

Neutrinos – little neutral particles created in huge numbers during nuclear reactions – were long thought to be massless. But during the 1980s

and 1990s the *solar neutrino problem* became undeniable: neutrino detection experiments on Earth were missing about half of the expected number of neutrinos made in the core of the sun during hydrogen burning. There were only two possibilities: either our model of the sun was way off[2] or something happened to the neutrinos on their way to us.

The correct explanation turned out to be something called *neutrino oscillations*, and the theory and subsequent observation of the effect earned the Nobel prize. In short, *electron* neutrinos are created in the core of the sun, but on their way to Earth some of them turn into *muon* neutrinos – don't worry too much about the names or why there's more than one kind of neutrino; the main idea is that neutrinos can oscillate back and forth between these two different types (actually, there's a *third* type, a tau neutrino, but we'll ignore that for simplicity).

As a slightly simplified model for neutrino mixing, we'll suppose the state of any neutrino – I'll call this the "flavour state," since particle physicists call the type of particle its flavour for some reason – can be written as a superposition of an electron neutrino state and a muon neutrino state:

$$|\nu\rangle = a|e\rangle + b|\mu\rangle. \tag{7.14}$$

This is a two-state system, something we've already studied in detail, so the physics of neutrinos is pretty much the same as spin-1/2 systems. To see how neutrinos oscillate, though, we'll have to calculate how this state evolves in time, meaning we'll need the Hamiltonian of the system.

The Hamiltonian, of course, is just the total energy, so we'll start with the relativistic energy of any particle (unfortunately, neutrinos are inherently relativistic objects – it's hard to slow them down),

$$E = \sqrt{p^2c^2 + m^2c^4}.$$

Although neutrinos were originally thought to be massless, it turns out we'll need to give them a very small mass in order for them to oscillate. That said, it will be a *very* small mass, so it's reasonable to think that their rest energy (mc^2) will be much smaller than their momentum (the pc term), in which case we can write

$$E = pc\left(1 + \frac{m^2c^4}{p^2c^2}\right)^{1/2} \approx pc\left(1 + \frac{m^2c^4}{2p^2c^2}\right),$$

[2] One of my undergraduate professors was a solar physicists, and according to him there was no way the solar models could be that wrong, so really there's only one possible explanation – something was wrong with our understanding of neutrinos.

or

$$E = pc + \frac{m^2c^3}{2p}. \tag{7.15}$$

For reasons we won't see until we solve the whole problem, I'm going to write the Hamiltonian as

$$\hat{H} \to \begin{pmatrix} pc + (m_1^2 + m_2^2)c^3/4p & (m_1^2 - m_2^2)c^3/4p \\ (m_1^2 - m_2^2)c^3/4p & pc + (m_1^2 + m_2^2)c^3/4p \end{pmatrix} \tag{7.16}$$

where m_1 and m_2 are two different masses. You might think that m_1 is the mass of the electron neutrino and m_2 is that of the muon neutrino, but that's not quite the case; hold tight and we'll see what that means.

Before we find the energy eigenvalues and eigenstates, consider for a moment what the Hamiltonian would look like if the neutrinos were in fact *massless*; then

$$\hat{H} \to \begin{pmatrix} pc & 0 \\ 0 & pc \end{pmatrix}.$$

This matrix is diagonal, so the possible energies are easy to read off: they're both the same, $E = pc$ as we'd expect for a massless relativistic particle. The eigenstates of the matrix are just the flavour basis states, $|e\rangle$ and $|\mu\rangle$, so the implication is that both neutrinos have the same energy. More importantly, though, the flavour states – which are also the energy eigenstates in this massless neutrino scenario – are stationary states, so neutrino oscillations simply wouldn't happen. Once an electron neutrino is created during a reaction, it will stay an electron neutrino forever.

Actually, it's possible to have massive neutrinos and still have the flavour states be stationary, as long as the masses are the same. In this case the matrix remains diagonal and we reach the same conclusions; evidently it's the *different* masses of the two neutrinos that lead to oscillations.

Okay, let's find the energies of the full Hamiltonian. For ease of writing stuff out, I'm going to write the Hamiltonian as

$$\hat{H} \to \begin{pmatrix} h & g \\ g & h \end{pmatrix},$$

where $h = pc + (m_1^2 + m_2^2)c^3/4p$ and $g = (m_1^2 - m_2^2)c^3/4p$. The characteristic equation is

$$\begin{vmatrix} h - E & g \\ g & h - E \end{vmatrix} = 0,$$

which has solutions

$$E_1 = h + g = pc + (m_1^2 + m_2^2)c^3/4p + (m_1^2 - m_2^2)c^3/4p = pc + m_1^2c^3/2p \tag{7.17}$$

and

$$E_2 = h - g = pc + (m_1^2 + m_2^2)c^3/4p - (m_1^2 - m_2^2)c^3/4p = pc + m_2^2 c^3/2p. \quad (7.18)$$

Now you can see why I chose the exact Hamiltonian I did – we get exactly the energies we expect based on equation 7.15. As long as the masses are different, these are two distinct energies.

Now that we have the energy eigenvalues, we can get the corresponding states as well. For $E = E_1 = h + g$, the eigenvalue equation is

$$\hat{H}|E_1\rangle = E_1|E_1\rangle,$$

or, in matrix notation,

$$\begin{pmatrix} h & g \\ g & h \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = (h + g) \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Solving gives $\beta = \alpha$, and after normalizing to find $\alpha = 1/\sqrt{2}$, the eigenstate is

$$|E_1\rangle = \frac{1}{\sqrt{2}}|e\rangle + \frac{1}{\sqrt{2}}|\mu\rangle. \quad (7.19)$$

Similarly, the eigenstate associated with energy $E_2 = h - g$ is

$$|E_2\rangle = \frac{1}{\sqrt{2}}|e\rangle - \frac{1}{\sqrt{2}}|\mu\rangle. \quad (7.20)$$

This is where things get interesting and a little weird. According to particle physics, a neutrino is created either as an electron neutrino (state $|e\rangle$) or as a muon neutrino (state $|\mu\rangle$). But the energy states are a superposition of these flavour states – evidently the electron neutrino, say, just doesn't *have* a definite energy. Put another way – because the energies depend on the two different masses – we can say the an electron neutrino does not have a definite mass; the m_1 isn't an m_{ν_e}. Alternatively, if we manage to measure the energy (or mass) of an electron neutrino, it's no longer an electron neutrino – its state collapses to one of the superpositions above and it doesn't have a definite flavour.

We can see how this leads to neutrino oscillations and solves the solar neutrino problem by supposing that, at the centre of the sun, an electron neutrino is created. The initial state is then

$$|\nu(0)\rangle = |e\rangle. \quad (7.21)$$

I think it's fair to call this neutrino an electron neutrino at this point.

To find the state at a later time, we need to first switch to the energy basis. For fun, I'll do that by multiplying the state by the identity operator:

$$|\nu(0)\rangle = \hat{1}|e\rangle = (|E_1\rangle\langle E_1| + |E_2\rangle\langle E_2|)\,|e\rangle.$$

Notice that I wrote the identity operator in the energy basis – this is the mechanism that switches basis for us. Working out the inner products gives

$$|\nu(0)\rangle = \frac{1}{\sqrt{2}}|E_1\rangle + \frac{1}{\sqrt{2}}|E_2\rangle \tag{7.22}$$

as you might expect; although it's fair to say this is an electron neutrino, we definitely can't say what the energy of the particle is – it simply doesn't have one.

Now we can tack on the exponential time factors to find the state at a later time in the usual way. We get

$$|\nu(t)\rangle = \frac{e^{-iE_1t/\hbar}}{\sqrt{2}}|E_1\rangle + \frac{e^{-iE_2t/\hbar}}{\sqrt{2}}|E_2\rangle.$$

Finally, I'll switch back to the flavour basis, although this time I'll do it using equations (7.19) and (7.20) directly rather than using the identity operator:

$$|\nu(t)\rangle = \frac{e^{-iE_1t/\hbar}}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}|e\rangle + \frac{1}{\sqrt{2}}|\mu\rangle\right) + \frac{e^{-iE_2t/\hbar}}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}|e\rangle - \frac{1}{\sqrt{2}}|\mu\rangle\right).$$

Using $E_1 = h + g$ and $E_2 = h - g$ and simplifying things, we get

$$|\nu(t)\rangle = \frac{1}{2}e^{-iht/\hbar}\left[\left(e^{-igt/\hbar} + e^{igt/\hbar}\right)|e\rangle + \left(e^{-igt/\hbar} - e^{igt/\hbar}\right)|\mu\rangle\right].$$

You might recognize cosine and sine in there, so we'll simplify further, but also note that there's an overall phase factor out in front – we'll drop that since it doesn't mean anything physically. Finally, the state at time t is

$$|\nu(t)\rangle = \cos(gt/\hbar)|e\rangle - i\sin(gt/\hbar)|\mu\rangle. \tag{7.23}$$

Take a look at what happens at time

$$\tau = \frac{\pi\hbar}{g} = \frac{\pi p\hbar}{4c^3}\frac{1}{m_1^2 - m_2^2};$$

at that point the cosine term goes to zero and the sine term becomes one and our state is

$$|\nu(\tau)\rangle = |\mu\rangle.$$

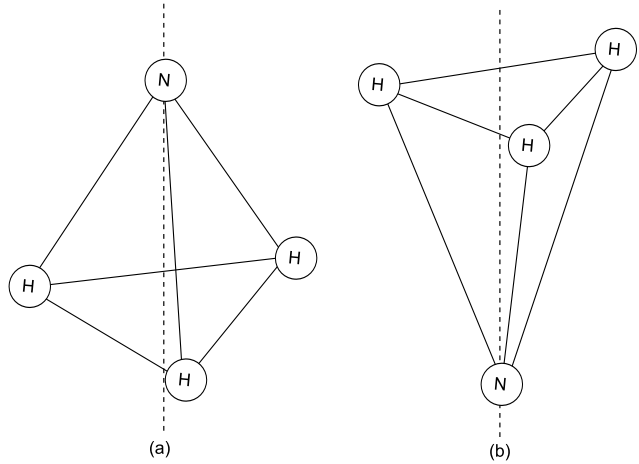


Figure 7.4: The nitrogen atom in ammonia can be found either above the plane (a) or below the plane (b) formed by the hydrogen atoms.

The electron neutrino has oscillated into a muon neutrino!

This is the explanation for the solar neutrino problem, although the full theory is a bit more complex since there are actually three different neutrino flavours. But the main idea is still the same – the flavour states are not the energy eigenstates, and are therefore not stationary – they instead oscillate amongst themselves.

Problems

Problem 7.1 – The Ammonia Molecule.

Ammonia (NH_3) is a complicated system of four nuclei and ten electrons interacting with each other to form bonds between atoms, making the stable state of the molecule a pyramid with three hydrogen atoms forming the base and a nitrogen atom at the apex – see Figure 7.4. Let's ignore all this internal dynamics and any translation or rotation the molecule may have. Because of the symmetry of the molecule, there are two places the N atom can reside: either above the plane formed by the hydrogen atoms or below the plane. This is, essentially, a two-state system.

We can describe the two states of the molecule by

$$|1\rangle = |\text{N above the plane}\rangle \quad \text{and} \quad |2\rangle = |\text{N below the plane}\rangle.$$

The Hamiltonian of the molecule in this basis is given by

$$\hat{H} \to \begin{pmatrix} E_0 & -A \\ -A & E_0 \end{pmatrix},$$

where A is a positive constant.

(a) Find the allowed energies $E_\pm$ and the associated energy eigenstates $|E_\pm\rangle$ for this system.

(b) Suppose the ammonia molecule is in the state $|E_+\rangle$. Where is the nitrogen molecule located in this state?

(c) At time $t = 0$ the molecule is in state $|1\rangle$. What is the probability of measuring the nitrogen atom to be *below* the plane at time t later?

(d) Ammonia gas clouds in galaxies can form a natural maser, which is like a laser but with microwave emission rather than visible light. The ammonia masers can be useful tracers of galactic dynamics among other things. The wavelength of the electromagnetic radiation from masers is 1.25 cm. Estimate the value of the constant A in the Hamiltonian (*Hint*: The microwave emission comes from photons that carry away the *difference* in energy between the two states, and recall that the energy of a photon is related to its wavelength from $E = hc/\lambda$, where $h = 6.626 \times 10^{-34}$ m^{2}kg/s.)

Problem 7.2 – Polarization of Light.

It is possible to polarize light in two different directions – let's say horizontal and vertical. Polarizing light is pretty easy to do – just pass it through a *polarizing filter* that is oriented either horizontally or vertically.[3] We won't worry about the properties of the electromagnetic waves, here, but instead focus on the quantum mechanics of polarized photons.

Let's suppose that a photon which is horizontally polarized is in state $|h\rangle$, and one which is vertically polarized is in state $|v\rangle$. In general, the photon can be in a superposition of both states,

$$|\gamma\rangle = a|h\rangle + b|v\rangle.$$

Experimentally, this is easy to do – just rotate the polarizing filter at an angle.

[3] If you're not familiar with polarizing filters, the lenses of some sunglasses are vertically polarized to block horizontally polarized light that results from reflections. Another application is in 3D movies – those glasses have one horizontal filter over one eye and a vertical filter over the other, so that each eye can see different images.

(a) Let's start with the basics. First pass unpolarized light, consisting of 100 photons, through a horizontal filter. (i) What state will the photons be in after that? (ii) Next, those photons are passed through a second horizontal filter. How many photons will pass through this filter? (iii) Finally, the photons are then passed through a vertical filter. How many photons will pass through this filter?

(b) Now we prepare a single photon in the state

$$|\gamma\rangle = |h\rangle - 2|v\rangle.$$

If you pass this photon through a horizontal filter, what is the probability that it will go through? What about through a vertical filter?

(c) It turns out that the polarization of a photon is related to its angular momentum **J**. In the $|h\rangle$-$|v\rangle$ basis, we can write the z-component of the angular momentum as

$$\hat{J}_z \to \begin{pmatrix} \hbar & 0 \\ 0 & -\hbar \end{pmatrix}.$$

(i) Is this operator *hermitian*? Do you expect it to be (and why or why not)?

(ii) What are the eigenvalues and eigenstates of $\hat{J}_z$?

(iii) If you were to make a measurement of J_z on a photon in the state $|\gamma\rangle$, what are the possible results you could measure, and what is the probability of getting each?

(iv) Calculate $\langle \hat{J}_z \rangle$ for the state $|\gamma\rangle$.

(d) In an experiment, a photon propagates along the optic axis (the x-axis in this case) of a quartz crystal. The Hamiltonian for this system, in the $|h\rangle$-$|v\rangle$ basis, is given by

$$\hat{H} \to \begin{pmatrix} 0 & -iE_0 \\ iE_0 & 0 \end{pmatrix}.$$

(i) Is this operator hermitian?

(ii) The operators $\hat{H}$ and $\hat{J}_z$ *don't* commute. Calculate their commutator

$$[\hat{H}, \hat{J}_z].$$

(iii) Since the Hamiltonian and z-component of angular momentum don't commute, they have an associated uncertainty principle. What is it (for the state $|\gamma\rangle$)? What does it imply about measurements of the energy and J_z?

(iv) Find the eigenvalues and eigenstates of the Hamiltonian.

(v) If you make a measurement of the energy of a photon in the state $|\gamma\rangle$, what are the possibilities and probabilities of each?

(e) A photon enters into the crystal, initially in state $|\psi(0)\rangle = |h\rangle$.

(i) What is the state of the photon a time t later, $|\psi(t)\rangle$? Write the state in the $|h\rangle$-$|v\rangle$ basis.

(ii) At time t, what is the probability of measuring the photon to be horizontally polarized? Vertically polarized?

Part II

Quantum Mechanics in One Dimension

Chapter 8

The Wave Function

8.1 One Dimensional Systems

Up until now we've dealt exclusively with *discrete* system like spin. In these cases, we used column vectors to represent the quantum state of our particle; for example, for spin-1/2, we had

$$|\psi\rangle \to \begin{pmatrix} \langle +|\psi\rangle \\ \langle -|\psi\rangle \end{pmatrix}.$$

Take careful note of the notation; each element in the column vector is a matrix element, and you can clearly see the basis $|+\rangle, |-\rangle$ used. In general, we would have a system that has n elements, where n could be two or three (say, for spin-1) or more – all the way up to infinity:

$$|\psi\rangle \to \begin{pmatrix} \langle a_1|\psi\rangle \\ \langle a_2|\psi\rangle \\ \langle a_3|\psi\rangle \\ \vdots \end{pmatrix}.$$

All of the quantum mechanical rules that we've discussed are identical regardless of the number of possible states of the system.

Now, though, it's time for something a little different (don't worry, only a little). We want to work in the *position representation*,[1] in which

[1] Sometimes I'll also call this the position basis, and sometimes position space; all three terms seem to be fairly common.

case we might write our state as the column vector

$$|\psi\rangle \rightarrow \begin{pmatrix} \langle x_1|\psi\rangle \\ \langle x_2|\psi\rangle \\ \langle x_3|\psi\rangle \\ \vdots \end{pmatrix}, \tag{8.1}$$

where the kets $|x_i\rangle$ are the position eigenstates. There's one problem here, though: not only are there an infinite number of position states, they are *continuous* – in general our particle could be at position $x = 1.0$, or $x = 1.01$, or $x = 1.000001$. In the language of linear algebra, we say that the spectrum of eigenvalues is continuous.

It's difficult to use a column vector to represent our state in the position basis – it's really more suited to discrete systems. But it's pretty easy to imagine how to extend the idea: each element in equation (8.1) is the state "evaluated" at the position x_i, and if these positions run continuously, what we really have is a *function*:

$$\boxed{|\psi\rangle \rightarrow \psi(x) \quad \text{where} \quad \psi(x) \equiv \langle x|\psi\rangle.} \tag{8.2}$$

We already know that we can represent a quantum state as a ket, or a column vector, and now we can also use a function. But what exactly does the function $\psi(x)$ mean? Well, remember that the inner product $\langle x|\psi\rangle$ is the probability *amplitude* (we have to square it to get the actual probability) of the quantum state to be measured in the position eigenstate $|x\rangle$. That means the probability of finding a particle to be at position x is

$$\mathscr{P}(x) = |\langle x|\psi\rangle|^2 = \langle\psi|x\rangle\langle x|\psi\rangle = \psi^*(x)\psi(x) = |\psi(x)|^2.$$

Actually, that's not quite correct; our jump to continuous systems means that $\mathscr{P}(x)$ is actually a probability *density*, and we properly have to say that $\mathscr{P}(x)dx$ is the probability of finding the particle between the position x and $x + dx$. If we want to know about a finite region, we'd have to integrate:

$$\boxed{\int_a^b \mathscr{P}(x)dx = \int_a^b |\psi(x)|^2 dx \quad \begin{array}{l}\text{is the probability of finding the} \\ \text{particle between } x = a \text{ and } x = \\ b.\end{array}} \tag{8.3}$$

Like all quantum states we've discussed so far, states written in the position representation must be normalized as well; here, that means

$$\langle\psi|\psi\rangle = 1 \quad \rightarrow \quad \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1. \tag{8.4}$$

In words, that means we have to find the particle *somewhere* with 100% probability.

The structure of the inner product above carries over to *all* inner products, so in general we'll calculate the inner product between state $|\phi\rangle$ and state $|\psi\rangle$ in position space as

$$\langle\phi|\psi\rangle \rightarrow \int_{-\infty}^{\infty} \phi^*(x)\psi(x)dx. \tag{8.5}$$

That means we can calculate probabilities in position space, too, with

$$\mathscr{P}_{\psi\rightarrow\phi} = |\langle\phi|\psi\rangle|^2 \rightarrow \left|\int_{-\infty}^{\infty} \phi^*(x)\psi(x)dx\right|^2 \tag{8.6}$$

being the probability of measuring the state $|\psi\rangle$ to be in the state $|\phi\rangle$.

What about expectation values? For that, we'll need to write the operator in position space as well:

$$\hat{A} \rightarrow A(x)$$

(as we'll see, operators in position space can take the form of functions of x or derivatives with respect to x). Then the expectation value carries over naturally to become

$$\langle A\rangle = \langle\psi|\hat{A}|\psi\rangle \rightarrow \int_{-\infty}^{\infty} \psi^*(x)A(x)\psi(x)dx. \tag{8.7}$$

For example, the expectation value of position itself, a quantity we'll be calculating frequently, is

$$\langle x\rangle = \langle\psi|\hat{x}|\psi\rangle \rightarrow \int_{-\infty}^{\infty} \psi^*(x)x\psi(x)dx = \int_{-\infty}^{\infty} x|\psi|^2dx. \tag{8.8}$$

8.2 Position Indeterminacy

We've already talked about indeterminacy – the idea that some observables simply don't have definite values in some quantum states – back when we talked about measuring spin in Section 3.3. But the issue hits a little differently when we add *position* into the mix – after all, position is something we readily understand on an everyday basis, whereas spin is some weird quantum thing.

Consider a particle described by the wave function $\psi(x)$ as shown in Figure 8.1. Even at a quick glance the wave function tells us quite a

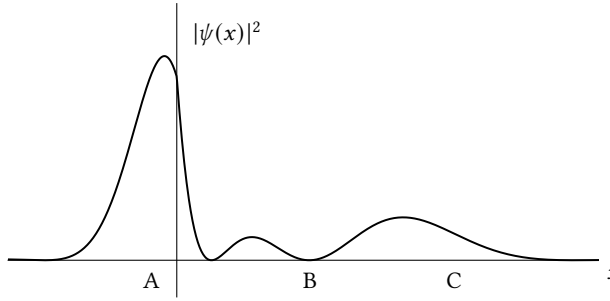


Figure 8.1: An example wave function.

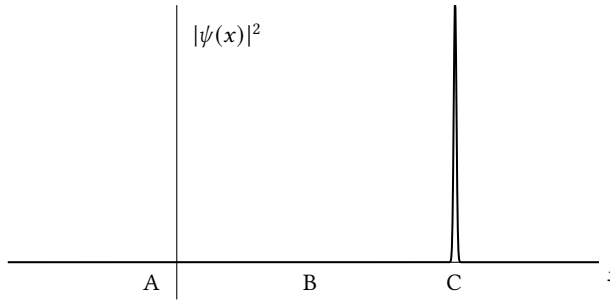


Figure 8.2: The wave function has collapsed to a spike at point C.

bit about where we would be likely to find the particle – more likely at position A, for example, and not at all at position B. Of course, like with spin, we can't say exactly where the particle is located *before* we measure its position – it's indeterminant. It doesn't just have an *unknown* position – it doesn't have a position at all.

To see this more clearly, suppose we go ahead and measure the position of our particle, and suppose that it happens to be found at position C. Once we make that measurement, the wave function will change – as with measuring spin, it will "collapse" to the eigenstate corresponding to the measurement we made. In this case that's a high, sharp spike at position C, as shown in Figure 8.2. Technically this is a Dirac delta function, as we'll discuss later, but it makes sense – subsequent measurements will keep returning position C, at least as long as they're made relatively quickly after the first and before the wave function evolves in time (which it will do according to the Schrödinger equation).

But then the question is – where was the particle *just before* its posi-

tion was measured? The options here are the same as back in Section 3.3: we could take a *realist* position (the particle was at position C), hold to the *Copenhagen interpretation* (the particle didn't have a position at all), or give up and refuse to answer. As before, we'll go with the standard Copenhagen interpretation, which says that the position of the particle was indeterminant – it somehow didn't have a position at all, and somehow measuring the particle caused it to "take a stand" and pick a position value consistent with the wave function of the particle. To say this is out of our normal range of experience is an understatement, but it seems to be how the universe works. The act of measurement in quantum mechanics is still a hotly debated and studied phenomenon among physicists.

8.3 The Energy Eigenvalue Equation

So, why are we talking about position all of sudden? Up to now we haven't had to worry about it – we discussed spin, or neutrino flavours, but haven't yet had to worry about *where* a particle is. But as you learned in your classical mechanics course, potential energy can depend on position, and, in one dimension, the total energy will be

$$E = T + V = \frac{p_x^2}{2m} + V(x),$$

where the first term is of course the kinetic energy (which depends on momentum) and the second is potential energy (which depends on position). In fact *any* dynamical variable can always be written in terms of momentum and position.

We should probably figure out how to deal with position and momentum in quantum mechanics, then. These are both *observables*, so to "turn on" quantum mechanics, we make them *operators*.

I'll start with position, which is a little easier to understand right away: the observable position x is represented by an operator $\hat{x}$. In the position representation, the operator just takes the form of x:

$$\hat{x} \to x. \tag{8.9}$$

That is, the actual operation is *multiply the state by the variable x*. This will all make sense when we use the operator, even if it sounds a little confusing now.

For the observable momentum p_x, the quantum operator is $\hat{p}_x$, and in the position representation

$$\hat{p}_x \to -i\hbar \frac{d}{dx}. \tag{8.10}$$

Observable	Operator	Position Representation	Operation
x	$\hat{x}$	x	Multiply by x
p_x	$\hat{p}_x$	$-i\hbar\frac{d}{dx}$	Take the derivative and multiply by $-i\hbar$

Table 8.1: The position and momentum observables in the position representation.

Somehow, the momentum operation is to take the derivative of the state and multiply by $-i\hbar$. These two observables are summarized in Table 8.1. And if you're wondering why the momentum operator is a spatial derivative, don't worry, we'll dig deeper into $\hat{p}$ later.

The Hamiltonian operator for a particle in one-dimensional motion along the x-axis can now be written down in position space as

$$\boxed{\hat{H} = \frac{\hat{p}_x^2}{2m} + V(\hat{x}) \rightarrow -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x).} \tag{8.11}$$

We normally write the energy eigenvalue equation as

$$\hat{H}|E_n\rangle = E_n|E_n\rangle,$$

but in the position representation we'll use a different letter ϕ to denote the energy eigenstate:

$$|E_n\rangle \rightarrow \phi_n(x).$$

The energy eigenvalue equation then becomes

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\phi_n}{dx^2} + V(x)\phi_n(x) = E_n\phi_n(x).} \tag{8.12}$$

This equation is often called the *time-independent Schrödinger equation*, although I'll keep calling it the energy eigenvalue equation. Notice that in position space we've gone from solving simple matrix equations to now solving differential equations.

8.4 Bound and Unbound States

Our goal for almost the remainder of the book will be to solve equation (8.12) – or its three dimensional version – for a variety of different potential energies $V(x)$. Obviously the solutions will depend pretty critically

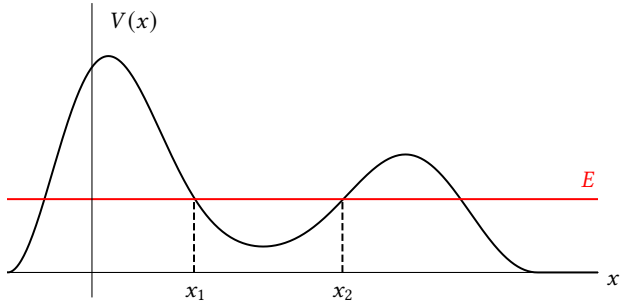


Figure 8.3: A generic example of a potential energy. For a particle of total energy E, the turning points are at x_1 and x_2.

on the exact form of $V(x)$, so it's worthwhile to remind you of how we deal with potential energy *classically* first.

Take a look at Figure 8.3, which shows a generic potential energy. If a particle has a total energy E – which will always be constant – as shown in red, then classically the particle is *bound* between the turning points at x_1 and x_2. The region outside the turning points is often called the *classical forbidden region*, and although in classical physics the particle will never be found outside the well, in quantum mechanics it is possible to *tunnel* through the barrier and escape, as we'll see later.

Classically, we distinguish between two types of motion, depending on the value of the energy. If $E < V_{\text{max}}$, as shown, then the motion is bound. However, if $E > V_{\text{max}}$, we call the motion *unbound* – although its speed might change as it passes through the potential, it will continue on to infinity and not be trapped.

In quantum mechanics there's an even greater distinction between bound and unbound states. We'll start our exploration of wave functions by looking at systems that only permit bound states, since they're a little simpler, and then turn to systems that have just unbound states or both bound and unbound together.

Problems

Problem 8.1 – Dealing with wave functions.

Consider the wave function

$$\psi(x) = Ae^{-\lambda|x|},$$

where λ is a positive constant.

(a) Normalize the wave function (that is, find the value of A such that $\psi(x)$ is normalized).

(b) Sketch $\psi(x)$ and $|\psi(x)|^2$.

(c) Calculate $\langle x \rangle$ and $\langle p \rangle$.

(d) Calculate $\langle x^2 \rangle$ and σ_x.

Problem 8.2 – Commutator of position and momentum.
Show that the commutator of $\hat{x}$ and $\hat{p}$ in position space is

$$[\hat{x}, \hat{p}] = i\hbar.$$

(*Hint:* they're operators, so it's useful to have something to operate on; use a "test function" $f(x)$.)

Problem 8.3 – Heisenberg's uncertainty principle.
Show that the position-momentum uncertainty principle is

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

Problem 8.4 – Normalization.
We learned back in Section 4.7 about the identity operator for spin; for a general discrete system we could write it as

$$\hat{1} = \sum_n |a_n\rangle\langle a_n|, \tag{8.13}$$

where the states $|a_n\rangle$ are the complete eigenstates of an operator. For a *continuous* system, though, we'd have to switch out the sum for an integral; for example, the position eigenstates form a complete set, so we can write

$$\hat{1} = \int |x\rangle\langle x| dx. \tag{8.14}$$

Show that by writing this identity operator between the bra and ket of the inner product in equation (8.4) you get the integral form of the normalization condition.

Chapter 9

The Infinite Square Well

9.1 The Particle in a Box

As our simplest example – a little *too* simple to serve as a good model for actual systems, but one that has all the essential features found in more complicated examples – consider a one dimensional "particle in a box." This particle of ours moves along the x axis only, and bounces perfectly elastically at wall that are located at $x = 0$ and $x = a$; that is, there's no loss of energy in the bounce. The particle is never allowed to break through a wall, no matter how fast it's going. And finally, the most essential feature: the particle remains at constant speed while in between the walls.

Can you picture this in your head? A good *classical* example of a particle in a box is a cart on an air track, with rubber bands at the ends allowing the cart to travel back and forth over and over again. With no friction in the air track, and perfectly elastic rubber bands, we have the same situation as above. Of, we're talking about quantum mechanics rather than classical – you can picture an electron bounding between two capacitors if you like – but I want to set up the problem using classical mechanics first.

Question: what would the potential energy of this system be? Well, we know it must be constant between the walls (so that there's no change in kinetic energy or the speed of the particle), and since absolute values don't matter to the potential energy, we might as well make it zero. What about at the walls? There we'll set the potential to be infinite – so that no matter how fast the particle is going it'll never have enough kinetic energy to make it over the walls and escape our potential well. Mathematically,

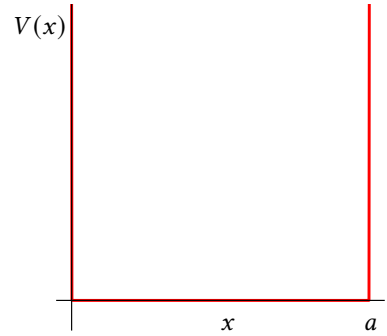


Figure 9.1: The potential energy for the infinite square well.

I'll write the potential as a piecewise function,

$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 < x < a \\ \infty, & x > a, \end{cases} \tag{9.1}$$

and it's plotted in Figure 9.1.

This system is called the infinite square well, and it's pretty easy to figure out a few things about the classical motion of a particle in the infinite square well. For example, between the walls the velocity is constant, and with $V(x) = 0$ there, a particle with total energy E would have velocity

$$v = \sqrt{\frac{2E}{m}}.$$

From this velocity and the knowledge that the particle has turning points at $x = 0$ and $x = a$, we could determine the position of the particle at any point in time we like, as long as we know it at one.

Of course, quantum mechanics won't allow us to predict exactly where the particle will be in the future, so if we want to compare with our quantum results later, we should instead think about probability. The probability density of the particle is inversely proportional to how fast it's going – the faster it moves, the less time it'll spend at that location. In general,

$$\mathscr{P}(x) \propto \frac{1}{v}, \tag{9.2}$$

but here the velocity is *constant*, so $\mathscr{P}(x)$ must be as well. Of course, it'll be *zero* outside of the box, and in order for the probability density to be

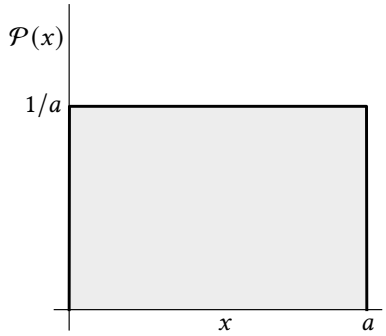


Figure 9.2: The classical probability density for the infinite square well. The shaded area must be equal to one – we must be able to find the particle in the box with 100% probability.

normalized, we must have

$$\mathcal{P} = \frac{1}{a}. \tag{9.3}$$

You can probably see this most easily in Figure 9.2.

9.2 The Allowed Energies and Wave Functions

To explore the quantum system, we need to solve the energy eigenvalue equation in position space,

$$-\frac{\hbar^2}{2m}\frac{d^2\phi_n}{dx^2} + V(x)\phi_n(x) = E_n\phi_n(x),$$

for the energy eigenvalues E_n as well as the energy eigenstates $\phi_n(x)$. For simplicity I'm going to drop the n label from the energy and state, and we'll add it in later.

Since the potential energy $V(x)$ is piecewise, we'll have to solve the differential equation in a piecewise manner, too. But for $x < 0$ or $x > a$, it's easy: we already know the particle will never be found outside the box, so we must have

$$\phi(x) = 0, \quad x < 0 \text{ or } x > a. \tag{9.4}$$

Remember this – we'll need it in a bit.

Inside the box ($0 < x < a$) the potential is zero, so the energy eigenvalue equation becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} = E\phi.$$

We're going to rewrite this a bit so that it becomes

$$\frac{d^2\phi}{dx^2} = -\frac{2mE}{\hbar^2},$$

and define a new constant k,

$$k \equiv \frac{\sqrt{2mE}}{\hbar}, \tag{9.5}$$

so that our differential equation becomes

$$\frac{d^2\phi}{dx^2} = -k^2\phi.$$

We'll see why we have k^2 in a moment; for now, keep in mind that k is both real and positive. That means the general solution to the equation is

$$\phi(x) = A\sin kx + B\cos kx$$

(this is a pretty common differential equation in physics, and I suggest you just memorize the solution; you'll need it frequently. Of course, if you don't believe me that this is in fact the general solution, go ahead and plug it in and check).

9.3 An Example

Problems

Problem 9.1 – To come.
To come

Chapter 10

The Finite Square Well

10.1 Bound States

What happens if the walls of our "box" weren't quite so high? Let's find out – suppose we have a particle in a *finite* square well, defined by the potential energy

$$V(x) = \begin{cases} V_0, & x < -a \\ 0, & -a < x < a \\ V_0, & x > a, \end{cases} \tag{10.1}$$

where V_0 is a positive constant – the depth of the well as shown in Figure 10.1. Notice one other difference from the infinite version as well: the well has been centered at $x = 0$ so the width of the well is actually $2a$; we'll see why in a little bit.

Actually, probably the biggest difference between the infinite square well and the finite well is what kind of states are allowed. Classically, for

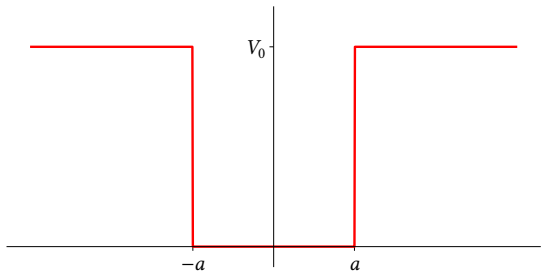


Figure 10.1: The potential energy of the finite square well.

the infinite square well, the particle in the box was always contained and could never escape, regardless of the energy of the particle. In the finite well, though, if $E > V_0$ the particle would not be trapped in the box and would instead escape to infinity. As we discussed back in Chapter 8, this case would be an *unbound state*, and unbound states have some additional complications in quantum mechanics – we'll save those for a later chapter. In the meantime, we'll only look for bound energy eigenstates, where

$$E < V_0 \quad \text{(bound states)},$$

but keep in mind that we'll be missing some of the eigenstates. One consequence of this is that the energy eigenstates we find here won't be *complete*.

Luckily, finding the bound states is pretty easy. For $x < -a$, to the left of the well, the energy eigenvalue equation reads

$$-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} + V_0\phi = E\phi,$$

or, cleaning it up a bit,

$$\frac{d^2\phi}{dx^2} = q^2\phi, \tag{10.2}$$

where

$$q = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{10.3}$$

Notice that q is always *real* since we're requiring that $E < V_0$. This should be a recognizable differential equation; the general solution looks like

$$\phi(x) = Ae^{qx} + Be^{-qx} \quad (x < -a). \tag{10.4}$$

Similarly, for $x > a$, the potential energy is again $V = V_0$ so our solution is the same, although we'll have to use different constants:

$$\phi(x) = Fe^{qx} + Ge^{-qx} \quad (x > a). \tag{10.5}$$

Finally, in the well itself we have $V = 0$, so the solution is the exact same as the infinite square well. We can write the energy eigenvalue equation as

$$\frac{d^2\phi}{dx^2} = -k^2\phi, \tag{10.6}$$

where

$$k = \frac{\sqrt{2mE}}{\hbar} \tag{10.7}$$

is once again real and positive. The general solution is

$$\phi(x) = C\sin(kx) + D\cos(kx) \quad (-a < x < a). \tag{10.8}$$

At the risk of repeating myself unnecessarily, I'll write out the full solution as a piecewise function; it helps to see it all together:

$$\phi(x) = \begin{cases} Ae^{qx} + Be^{-qx}, & x < -a \\ C\sin(kx) + D\cos(kx), & -a < x < a \\ Fe^{qx} + Ge^{-qx}, & x > a. \end{cases} \tag{10.9}$$

Now all we have to do is find all six constants A, B, C, D, F, and G, plus of course our allowed energy E, which is buried inside q and k. To do that we use *boundary conditions* as usual.

10.2 Boundary Conditions

Part III

Quantum Mechanics in Higher Dimensions